SonaDx – Early Breast Cancer Detection from Sonography Images

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Declaration

This is to declare that the thesis work entitled "**SonaDx – Early Breast Cancer Detection from Sonography**" has been carried out by **Md. Shakib Ahamed, Md. Tanveer Ahammed Tarek, Md. Omayer Hasan Muhit, Rawnak Tasnim Ruku** in the Department of Computer Science and Engineering (CSE), University of Information Technology and Sciences (UITS), Dhaka, Bangladesh. The above thesis work or any part of this work has not been submitted anywhere for the award of any degree or diploma.

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Justification Report

Below is the CO-PO Mapping and Complex Engineering Problem (CEP) Justification Report for the project titled "SonaDx – Early Breast Cancer Detection from Sonography Images"

CO-PO Mapping:

CO No.	Course Outcome Description	Associated PO	Knowledge Profile (KP)
CO1	Apply engineering fundamentals and knowledge to identify and translate a real-life problem into a viable engineering and/or computing solution.	PO1:Engineering Knowledge	K3 (Engineering fundamentals)
CO2	Identify, formulate, and analyze the specific outcomes and functional requirements of the proposed solution to reach substantiated conclusions.	PO2:Problem Analysis	K2 (Mathematics & analytical methods)
CO3	Design and build complex engineering systems, components, or processes that meet identified functional requirements and specified needs.	PO3: Design/ Development of Solutions	K5 (Engineering design knowledge)
CO4	Select and apply appropriate modern engineering software or hardware tools required for the analysis and design of a system or subsystem.	PO5: Modern Tool Usage	K6 (Engineering tools and practice)
CO5	Function effectively as an individual and as a member or leader in a multidisciplinary team.	PO9: Individual and Team Work	N/A (Professional skill)
CO6	Demonstrate knowledge of project management principles by preparing a timeline, formulating an appropriate budget, and identifying the economic impact of project considerations.	PO11: Project Management & Finance	N/A (Professional skill)
CO7	Conduct investigations by analyzing, evaluating, and validating system performance against specifications using research methods and appropriate tools.	PO4: Investigation	K8 (Research-based knowledge)
CO8	Assess the societal, health, safety, legal, and cultural impacts and responsibilities of the design project using contextual knowledge.	PO6: The Engineer and Society	K7 (Engineering in society)
CO9	Understand and evaluate the environmental and sustainability impacts of the design project in the solution of complex	PO7: Environment & Sustainability	K7 (Engineering in society)

	engineering problems.		
CO10	Assess the professional ethics and responsibilities and norms of engineering practice related to the design project.	PO8: Ethics	K7 (Engineering in society)
CO11	Communicate effectively by presenting design project results through written technical documents and oral presentations.	PO10:Communication	N/A (Professional skill)
CO12	Recognize the need for, and engage in, independent life-long learning by seeking out and utilizing appropriate resources needed to validate the proposed engineering solution.	PO12: Lifelong Learning	N/A (Professional skill)

Justification of PO through COs:

CO1 (PO1) : Real-Life Problem Identification: The project addresses the real-world problem of delayed breast cancer diagnosis in resource-limited settings by proposing a web-based solution for early screening using sonography images. Engineering and computing fundamentals were applied to translate a healthcare problem into a functional software system. For example, the team applied knowledge of biomedical imaging and computer science basics to frame the problem and scope the solution.

CO2 (PO2) : Requirement Analysis: Functional and non-functional requirements of SonaDx were clearly identified, formulated, and analyzed to ensure the system meets user and clinical needs. The team determined key outcomes such as image upload and preprocessing, tumor classification accuracy, diagnosis report generation, and user role management. By thoroughly analyzing requirements (e.g. need for high sensitivity and specificity, fast response time, and data privacy), the project reached substantiated conclusions about system design choices.

CO3 (PO3) : System Design and Development: SonaDx was designed and implemented as a complete system integrating a deep learning model, image processing pipeline, backend server logic, and a user-friendly frontend interface to fulfill the identified requirements.The design process involved creating a CNN-based classification model(EfficientNetB3) for tumor detection and incorporating explainable AI features(Grad-CAM heatmaps) for interpretability.The team followed an iterative design approach, ensuring the final system architecture meets specified needs in terms of accuracy,usability,and scalability.

CO4 (PO5): Use of Modern Tools: Modern software tools and frameworks were selected and applied for efficient development and deployment of the solution.The team utilized state-of-the-art libraries and platforms such as Python with TensorFlow/Keras for deep learning model training (EfficientNetB3),OpenCV for image processing,and Flask/Django for the web application backend.They also leveraged technologies like TensorFlow Lite for model deployment on modest hardware and used version control and collaboration tools (e.g. GitHub) to manage the project.This appropriate use of modern engineering tools ensured the

analysis,design,and implementation were done efficiently and in line with current professional practices.

CO5 (PO9): Teamwork:The project was completed by a group of four members who collaborated effectively,each contributing in a multidisciplinary team role.Team members functioned as individuals and as a coordinated unit,sharing responsibilities across development,testing, documentation, and integration tasks. Regular team meetings, division of labor (e.g. one focusing on the model, another on frontend UI, etc.),and use of collaboration platforms enabled the group to work cohesively.This teamwork mirrors a real engineering environment where communication and coordination among software developers,data scientists,and domain experts (such as medical professionals) were crucial to the project's success.

CO6 (PO11): Project Management:Sound project management principles were demonstrated through careful planning and resource management. A structured timeline was developed and followed to complete the project by the target deadline (20 January 2026), with milestones for data collection, model development,testing, and deployment.The team formulated an appropriate budget (for computing resources, data acquisition, and any cloud services) and managed these resources efficiently. They also considered the economic aspects of the solution,ensuring the proposed system would be cost-effective (e.g. using open-source tools and commodity hardware) and viable for deployment in low-resource healthcare settings.

CO7 (PO4): Investigation and Validation:The system's performance was thoroughly investigated and validated using scientific methods and appropriate tools. The team conducted experiments to evaluate the model's accuracy,precision, recall,and F1-score on test data, and analyzed confusion matrices to understand misclassification rates.They applied research methods such as cross-validation and hyperparameter tuning to improve model generalization. The SonaDx system was validated against expected outcomes and existing literature benchmarks,ensuring that results are reliable for diagnostic support. This investigative approach, which included exploring explainability(e.g. verifying that Grad-CAM highlights actual tumor regions),confirms that the solution meets its performance specifications.

CO8 (PO6): Societal, Health, and Safety Impact:The project considers broad societal and health impacts by promoting early breast cancer detection in the community.Using contextual knowledge of public health and patient concerns,the team addressed safety and legal responsibilities:the system is designed as a decision support tool rather than a definitive diagnostic,and it clearly communicates that its outputs are preliminary screenings not a final medical diagnosis.The solution has cultural and societal sensitivity in mind, being a web-based tool that can be accessed in remote clinics or at home,thus potentially reaching underserved populations.Legal and privacy considerations (like securing patient data and complying with relevant health data regulations)were also taken into account to ensure responsible deployment.

CO9 (PO7): Environmental and Sustainability Considerations:The SonaDx solution supports environmental sustainability and efficient use of resources.By enabling digital ultrasound screening and telemedicine features,the system can reduce the need for unnecessary patient travel to distant hospitals,thereby cutting down on associated fuel use and emissions. The project leveraged efficient computational models (using a lightweight CNN architecture and optimized code) that require relatively modest hardware,which translates to lower energy consumption in deployment.The focus on a software-driven diagnostic aid also means less reliance on disposable physical resources(like film prints of scans or paperwork), aligning the project with sustainable and eco-friendly practices.

CO10 (PO8): Ethics and Professional Responsibility:The team upheld professional ethics and responsibility norms throughout the project.They ensured patient data privacy by anonymizing all sonography images and securing any sensitive information, following ethical approval protocols for using medical data.The developed model and website include disclaimers to avoid misleading claims about diagnostic certainty,emphasizing that the tool is to aid (not replace)clinical judgment.The engineers practiced integrity by accurately reporting results (including limitations of the AI model) and made design choices that prioritize patient well-being and safety. In essence,the project adhered to the standards of ethical engineering practice in healthcare technology.

CO11 (PO10): Communication:Project outcomes and learnings were communicated effectively to both technical and non-technical audiences.The team produced comprehensive written documentation, including a detailed thesis report and this justification report,which explain the system design, experimental results, and the significance of the work.They also prepared visual diagrams (such as use case diagrams,workflow charts, and system architecture schematics) to illustrate the project structure and data flows clearly.Throughout the capstone,the team delivered oral presentations and demonstrations (for instance,interim progress presentations and a final defense) using clear technical language and effective presentation tools. These communication efforts ensured that the significance and functioning of SonaDx were well understood by evaluators, medical collaborators, and peers.

CO12 (PO12) : Lifelong Learning:The project required the team to independently learn and adapt to new technologies and domain knowledge,highlighting the importance of continuous self-directed learning.Team members engaged with research literature on deep learning in medical imaging, studied advanced techniques like explainable AI (Grad-CAM,LRP, etc.), and learned to use new development tools (such as unfamiliar libraries and cloud deployment platforms).They also updated their knowledge on breast ultrasound interpretation by consulting academic papers and, where possible, professionals in the medical field.This process of seeking out and utilizing new resources to validate and improve the engineering solution exemplifies a commitment to lifelong learning,ensuring that their skills and understanding remain current in a rapidly evolving field.

Complex Engineering Problem(CEP) Justification:

The capstone project was a Complex Engineering Problem as evidenced by the following attributes (P1–P7):

- **P1**:Depth of Knowledge Required –The problem could not be resolved with basic knowledge alone; it required advanced,in depth engineering knowledge.The team needed to apply and integrate concepts beyond the undergraduate curriculum, including specialized machine learning techniques for image analysis(e.g. fine-tuning an EfficientNetB3 CNN model),software architecture for a full stack web application, and understanding medical data handling practices.This breadth and depth of knowledge (spanning theory and application)were essential to approach the challenge from first principles and achieve a viable solution.
- **P2:** Range of Conflicting Requirements The project involved wide ranging requirements that had to be balanced against each other. For example,achieving high diagnostic accuracy and sensitivity conflicted with the need for fast,real-time performance and a smooth user experience.Similarly,maximizing model complexity for better predictions had to be weighed against limitations in computation resources for deployment.The team also balanced usability for doctors and patients with strict data security/privacy requirements.These conflicting technical and stakeholder requirements necessitated trade-off analysis and creative compromises in the final design.
- **P3**: Depth of Analysis Required There was no obvious,off-the-shelf solution to the problem,requiring originality and significant analysis to formulate a suitable approach.The team engaged in abstract thinking and experimentation to model the problem: they explored various CNN architectures (MobileNet,Xception, EfficientNet, etc.),conducted extensive parameter tuning,and analyzed results to iteratively improve the model.The absence of a clear solution path meant the students had to critically analyze data patterns and model behaviors to discover an effective strategy for early tumor detection,embodying a high level of analytical depth.
- **P4**: Familiarity of Issues Many issues tackled in this project were unfamiliar and infrequently encountered in typical coursework.The domain of medical ultrasound imaging and interpretation is not a routine area in undergraduate projects,so the team had to research new subject matter (e.g. characteristics of ultrasound images,common artifacts, and how tumors appear in sonography).Integrating an AI solution into a healthcare context (with considerations like patient data ethics and doctor workflows) was also novel for the team.This unfamiliarity required significant self-study, consultation with domain experts,and adaptation of knowledge from other fields, highlighting the complexity of the problem context.
- **P5**: Codes and Standards Applicability The problem lay partially outside the scope of standard engineering codes or ready made guidelines,especially since AI in healthcare is a frontier area.There is no specific engineering textbook procedure or established standard that directly guided the creation of an ultrasound-based diagnostic AI with XAI features.Thus, the team had to rely on independent judgment, adopting best practices from research literature and analogous standards (such as general software engineering

principles or medical device software guidelines) where applicable. They often had to make design decisions in uncharted territory for instance, determining acceptable performance thresholds for a screening tool which underscores the complexity beyond straightforward, codified problems.

- **P6**: Extent of Stakeholder Involvement – The project involved diverse stakeholders with potentially conflicting needs. Radiologists and clinicians require a tool that is accurate and trustworthy (with explanations for predictions), patients seek reassurance and privacy, and the university evaluators expected a robust technical implementation. Additionally, hospital administration (for the local data) had concerns regarding ethical data use. The team had to consider input from and impact on these different groups. Balancing stakeholder expectations such as making the interface both clinician-friendly and patient-accessible, or ensuring the AI's recommendations respect medical professionals' judgment added to the problem's complexity and required a holistic, stakeholder-conscious approach.
- **P7**: Interdependence (Complexity of Sub-Problems) The overall problem was high-level and comprised many interrelated sub-problems and components.The solution needed the successful integration of multiple subsystems:data collection and preprocessing,neural network training,explainability module, backend server,web frontend, and deployment infrastructure.Each component (e.g., the image classification engine vs. the web interface) could be seen as a significant problem on its own,yet all had to function together seamlessly.A change or issue in one part (such as model output format) could directly impact other parts (like how the frontend displays results),reflecting a tightly interdependent system.Managing these interconnections and ensuring compatibility across components was a complex task requiring careful coordination and systems thinking.

By addressing P1–P7,the project clearly meets the criteria of a complex engineering problem: it required advanced knowledge and analysis,involved multifaceted and novel issues with no simple solution,needed consideration of various stakeholder needs,and comprised numerous interlinked technical components.

Range of Complex Engineering Acrivities (CEA) Justification:

The activities carried out in this capstone project correspond to Complex Engineering Activities, particularly demonstrating attributes A1, A3, and A4:

- **A1:** Range of Resources –The project made use of diverse resources,both human and material.It involved a team of four people working in various roles (data scientist, developer, etc.),utilizing knowledge from multiple disciplines (computer vision, software engineering,biomedical informatics).The team leveraged a variety of equipment and technology resources: a curated dataset of 2,644 ultrasound images from different sources,high-performance computing resources (for model training and experimentation) ,and cloud based tools for deployment testing.They also managed informational resources like research papers,documentation,and consultation with

domain experts.This wide range of people, data ,tools,and knowledge bases underscores the complexity of the engineering activity.

- **A3:** Innovation –The project required creative and innovative application of engineering principles and research-based knowledge. Rather than applying a standard solution,the team innovated by customizing and fine-tuning a state-of-the-art deep learning architecture (EfficientNetB3)for their specific problem.They introduced a novel hybrid dataset approach (combining global and local data) to improve model generalization, which is a creative solution to data scarcity.Furthermore integrating an explainable AI mechanism (Grad-CAM heatmaps) into the diagnostic workflow exemplifies innovative thinking to increase the system's trustworthiness.These creative measures show that the project's activities were not routine they pushed the boundaries of typical undergraduate work by bringing research concepts into practical implementation in new ways.
- **A4:** Consequences for Society and the Environment – The engineering activities were conducted with a keen awareness of their societal and environmental consequences. On the societal side, this project's outcome directly contributes to public health: by facilitating early cancer detection, it has the potential to save lives and improve outcomes in the community. The team recognized the responsibility this entails, ensuring accuracy and reliability to avoid harm from false results and striving to make the technology accessible (for example designing the interface for ease of use by healthcare workers in clinics). Regarding environmental impact the project favored digital solutions that reduce the need for physical resources for instance digital record-keeping and online consultations can cut down on paper use and travel, as mentioned earlier.While the primary focus was healthcare, the solution's efficient design (lightweight model and web deployment) also means lower energy consumption and hardware requirements, which is beneficial for the environment. In summary,the project's activities and design decisions took into account the broader impact on society's well-being and strove to minimize any negative environmental footprint.

Mapping to PO10 (Communication):

Effective communication(Program Outcome 10)was an integral part of the capstone project. The team ensured that their work was well documented and presented clearly through various forms:

- Written Technical Documents:Preparation of a comprehensive project thesis,this justification report,and regular progress reports.These documents detailed the system design, methodologies,results,and analysis in a clear and organized manner for academic and professional audiences.
- Oral Presentations:Delivery of formal presentations during project defense and interim evaluations.The team explained technical concepts (such as the CNN model architecture and results)in an accessible way,answered questions from examiners,and incorporated feedback.They also demonstrated the working prototype to stakeholders, showcasing effective speaking and demonstration skills.

- Visual Representation:Use of diagrams and visual aids to communicate complex information. This included system architecture diagrams illustrating the endtoend workflow (from image input to diagnosis output),use case and sequence diagrams to depict system functionality,and Grad-CAM heatmap visualizations to explain the model's predictions.Such visual elements helped convey the project's design and findings intuitively to diverse audiences.

Through these communication efforts,the team successfully conveyed their engineering work. They adhered to professional standards in technical writing and presentation,ensuring that SonaDx's objectives, processes, and results were understood and appreciated by others in the engineering community and the healthcare field.The emphasis on clear communication not only fulfills PO10 but also helped in multidisciplinary collaboration, as the project bridged computer engineering and medical domains.

Abstract

Breast cancer is one of the leading causes of mortality among women worldwide, and early detection remains the most effective strategy for improving survival outcomes. In developing regions such as Bangladesh, limited access to expert radiologists and advanced imaging technologies often delays diagnosis. This study presents SonaDx, a deep learning–based breast cancer detection system that leverages ultrasound imaging for affordable, accessible, and accurate screening. A hybrid dataset of 2,644 ultrasound images integrating BUSI, MT Small, Breast-Lesions-USG, Kaggle, and local Bangladeshi samples are curated and preprocessed for model training. The proposed model is based on EfficientNetB3, fine-tuned using transfer learning to classify images into Normal, Benign, and Malignant categories. The system has achieved 94.5% testing accuracy, demonstrating strong generalization across diverse imaging conditions. Grad-CAM visualizations are incorporated for interpretability, while the model is converted to TensorFlow Lite for lightweight deployment via a Flask API and integrated with a React web interface hosted on Hugging Face and Cloudflare. The results confirm that SonaDx provides high diagnostic accuracy, fast inference, and real-time visualization, bridging the gap between research and clinical application. This work underscores the potential of AI-driven ultrasound analysis to support early breast cancer detection in resource-limited healthcare settings.

Keywords: Breast Cancer Detection, Deep Learning, Ultrasound Imaging, EfficientNetB3, Transfer Learning, Tensor Flow Lite, Mobile Health Application, SonaDx.

Preface

This B.Sc. The capstone project is outlined based on the results obtained from model training and testing. This is carried out in the Department of Computer Science and Engineering (CSE), Faculty of Science and Engineering, at University of Information Technology and Sciences (UITS), Dhaka, Bangladesh.

This capstone project includes 6 chapters which are briefed as follows:

Chapter 1: Introduction

1.1 Introduction
Provides an overview of breast cancer's global and local impact, emphasizing the need for AI-assisted ultrasound diagnosis in Bangladesh.
1.2 Motivation
Explains the rationale behind developing an accessible deep learning system to support early detection and overcome radiologist shortages.
1.3 Aims & Objectives
States the project's primary aim to build a CNN-based model for breast cancer detection with defined objectives for data, training, and deployment.
1.4 Challenges
Outlines difficulties in data quality, model generalization, interpretability, and deployment within low-resource healthcare settings.
1.5 Contribution
Summarizes the novel dataset creation, model development, evaluation, and interface implementation achieved in this study.
1.6 Consummation (Summary)
Concludes with a roadmap linking all chapters and reiterates the project's relevance to medical AI in Bangladesh.

Chapter 2: Background Studies

2.1 Introduction: Reviews the importance of deep learning in medical imaging, focusing on CNNs for breast ultrasound analysis.
2.2 Literature Review
Summarizes key research on CNN architectures and datasets like BUSI, highlighting trends and findings.
2.3 Related Work
Discusses prior efforts and their limitations in dataset diversity and model generalization.
2.4 Gaps and Challenges
Identifies data scarcity, image quality issues, overfitting, and limited model explainability as key challenges.
2.5 How This Project Builds on Existing Work
Describes SonaDx's hybrid dataset, refined preprocessing, and lightweight deployable model.

2.6 Corollary
Concludes that localized, explainable AI models are vital for real-world diagnostic success.

Chapter 3: Methodology

3.1 Introduction
Provides an overview of the methodological framework adopted for designing, training, and evaluating the proposed SonaDx system.
3.2 System Workflow
Illustrates the overall workflow of the system, showing the sequence from image input to prediction output and visualization.
3.3 Dataset Collection
Describes the sources, class distribution, and composition of breast ultrasound images used in this study.
3.4 Data Preprocessing
Explains the preprocessing techniques applied to improve image quality and ensure compatibility with deep learning models.
3.5 Model Development
Details the design and selection of convolutional neural network architectures used for breast cancer classification.
3.6 Training and Testing
Outlines the training strategy, data splitting, and testing procedures used to evaluate model performance.
3.7 Explainability Using Grad-CAM
Introduces the Grad-CAM technique used to visualize model decision regions and enhance interpretability.
3.8 Evaluation Metrics
Defines the performance metrics employed to assess classification accuracy and model reliability.
3.9 Consummation (Summary)
Summarizes the complete methodological approach and prepares the foundation for result analysis in the next chapter.

Chapter 4: Implementation

4.1 Introduction
Introduces SonaDx's implementation approach using EfficientNetB3 integrated within a web-based system.
4.2 System Architecture
Explains modular design across data, model, backend, frontend, and cloud layers.
4.3 Backend Development
Details Flask API development, image preprocessing, and inference workflow.
4.4 Frontend Development
Describes the React-based interface that enables real-time user interaction and visualization.

4.5 Deployment and Integration
Covers model conversion to TensorFlow Lite and cloud deployment using Hugging Face and Cloudflare.

Chapter 5: Result Analysis

5.1 Introduction
Introduces the purpose of result analysis and explains how the experimental outcomes and comparisons are organized in this chapter.
5.2 Model Selection and Comparison
Compares the performance of different CNN architectures to justify the selection of the most effective model for breast cancer detection.
5.3 Evaluation Metrics and Efficiency
Analyzes model performance using standard evaluation metrics and discusses computational efficiency and resource utilization.
5.4 Grad-CAM Visualization and Interface Integration
Demonstrates the use of Grad-CAM heatmaps and their integration into the application interface to enhance interpretability and user trust.
5.5 Real-World Applicability
Discusses the practical deployment potential of the proposed system in real healthcare settings, considering usability and clinical relevance.
5.6 Conviction (Summary of Findings)
Summarizes key findings from the result analysis and highlights the overall effectiveness and limitations of the proposed approach.

Chapter 6 : Mobile Application Design for SonaDx

6.1 Introduction
Introduces the important of mmobile/web applications in extendinig AI-based breast cancer diagnostic services to underserved regions of Bangladesh.
6.2 Application Objectives
Defines the key objectives of SonaDx, focusing on real-time diagnosis, accessibility, usability, security, and scalability.
6.3 Key Features
Summarizes the core functionalities of the app, including image upload, AI-based classification, explainable heatmap visualization, and secure result delivery.
6.4 Technical Overview
Briefly describes the application's technical architecture, covering front-end and back-end technologies ,ML integration, and cloud deployment.
6.5 Use Case Diagram
Outlines the primary actors and interactions of the system, illustrating how users upload images and receive diagnostic results.
6.6 App Workflow Diagram
Explains the sequential workflow of the application from image submission to AI analysis and result presentation.

6.7 App Interface Demonstration
Highlights the user-friendly, responsive, and minimal interface design optimized for both mobile and desktop platforms.
6.8 Practical Benefits
Discusses the real-world advantages of SonaDx, including early detection support, assistance for rural clinicians, and reduced diagnostic delays.
6.9 Future Plans
Summarizes proposed enhancements as offline mobile support, Bangla language integration, and expansion to multi model diagnostics.
6.10 Picture of the UI
Presents the graphical user interface of the SonaDx application, demonstrating the layout of the home page, image upload section, and result visualization screen.
6.11 Consummation (Summary)
Concludes the chapter by emphasizing how the application design bridges research and real world deployment, reinforcing the contribution of SonaDx to AI-driven breast cancer screening in Bangladesh.

Chapter 7: Conclusion and Future Work

7.1 Conclusion
Summarizes the success of SonaDx in achieving high accuracy and real-world deploy ability.
7.2 Future Works
Suggests improvements in dataset expansion, explainability, mobile deployment, and clinical validation.
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Reflects on the societal impact and long-term potential of AI in advancing medical diagnostics.

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Abbreviations and Symbols

AI	**Artificial Intelligence**
DL	**Deep Learning**
CNN	**Convolutional Neural Network**
MRI	**Magnetic Resonance Imaging**
CSS	**Cascading Style Sheet**
BUSI	**Breast Ultrasound Images**
API	**Application Programming Interface**
JSON	**JavaScript Object Notation**
UI	**User Interface**
CPU	**Central Processing Unit**
GPU	**Graphics Processing Unit**
NPU	**Neural Processing Unit**
ROC	**Receiver Operating Characteristic**
AUC	**Area Under the Curve**
LRP	**Layer-Wise Relevance Propagation**
ViT	**Vision Transformer**
Grad-CAM	**Gradient-Weighted Class Activation Mapping**
mHealth	**Mobile Health**
BI-RADS	**Breast Imaging-Reporting and Data System**
CDN	**Content Delivery Network**
HTTPS	**HyperText Transfer Protocol Secure**
%	**Percent**
±	**Plus - Minus**
≈	**Approximately Equal**
σ	**Sigma-Standard Deviation**

Chapter 1

Introduction

1.1 Introduction

Breast cancer is still one of the leading global causes of cancer morbidity and mortality among women today [1].It's critical to find and correctly classify at an early stage: The evidence is that where cancers are caught in the early grades Interventions are more effective and less invasive, increases the rate of survival improves without greatly effecting the quality of life. Worse still in the country there is little sophisticated imaging equipment and diagnostic services are not readily available, especially in rural areas and towns in the periphery of larger urban centers where most people actually live. There is a shortage of trained radiologists, aggravating matters still further [2].

Artificial intelligence (AI) and its application of deep learning have been increasingly applied in medical imaging tasks from analysis of images to practical tools which even the human visualist can utilize. Among these, Convolutional Neural Networks (CNNs) stand out because they have been so effective in the detection, classification, and segmentation of images. They can : automatically learn hierarchical features (edges, textures, patterns, shapes) from raw image data-hence the need for hand-crafted features (based on expert knowledge of a particular field) may be eliminated. CNNs were widely used in breast ultrasound, mammography, MRI, and histopathology across many studies with impressive results [3]. For example, in the field of breast ultrasound to cite one instance, deep learning methods have displayed improvements in diagnosis performance while at the same time significantly reducing reliance upon operator experience and increasing epidemiological reliability.

In this paper we collected 2,644 breast image samples (partial dataset from Kaggle, and the rest from Bangladeshi hospitals) and designed a number of CNN architectures. We achieved an accuracy of 94.5% on testing. This indicates strong potential for a local in-depth AI-assisted diagnostic tool. In this thesis, we introduce the motivation, methodology, evaluation, and challenges; and hope to lay a path toward implementation in Bangladeshi medical settings.

1.2 Motivation

The motivation for this research is multifaceted:

■ **Local health need and late diagnosis:** In Bangladesh many breast cancer cases are detected late, when treatment is more difficult and survival rates lower [4]. A reliable automated screening tool can help detect suspicious cases earlier, especially in underserved areas.

■ **Scarcity of expert readers**: Radiologists and specialists are concentrated in urban centers. Many smaller hospitals and diagnostic labs have limited access to expert interpretation. (62% rural population yet 75% doctors serve in urban areas, 2025) AI tools can provide second opinions or screening support [5].

■ **Variation in imaging conditions**: Images collected in Bangladesh (equipment, protocols, patient demographics) may differ from standard datasets used globally. Training on local data ensures better performance in the target population [6].
■ **Performance potential of CNNs**: Many recent studies have shown that CNNs, particularly when combined with transfer learning, ensemble methods, or hybrid designs, can deliver high accuracy in breast cancer image classification (e.g. accuracies nearing 94.5% in ultrasound contexts) [7].
■ **Scalability & affordability**: Once a model is trained and validated, deployment (on servers, edge devices, or cloud) can scale to many facilities at relatively low marginal cost. This is vital in resource-limited settings.

1.3 Aims & Objectives

Aim

To develop and validate CNN-based machine learning models for breast cancer detection using imaging data collected from Bangladeshi hospitals, achieving robust performance (testing accuracy ~94.5%) and a user interface prototype suitable for clinical usage.

Objectives

1. Collect and curate a dataset of 2,644 breast imaging samples (benign, malignant, possibly normal) from Bangladeshi hospitals, with appropriate ethical approvals and anonymization.
2. Preprocess images (resizing, normalization, augmentation, denoising) to standardize inputs and mitigate noise or artifacts.
3. Design and train various CNN architectures (MobileNetV2, Xception, EfficientNetB3) to classify images.
4. Perform hyperparameter tuning, cross-validation [8], and model comparison to choose best-performing models.
5. Achieve a testing accuracy of ~94.5%; compute additional metrics (precision, recall, specificity, F1-score, ROC-AUC) to fully evaluate performance.
6. Develop a prototype user interface (web app) through which clinicians can upload images, view predictions, and inspect interpretability maps (e.g. heatmaps, Grad-CAM).
7. Assess generalization and robustness by testing on held-out subsets, cross-hospital validation, or data from different imaging devices.
8. Document challenges, limitations, and deployment pathways for integration into the Bangladeshi healthcare infrastructure.

1.4 Challenges

Implementing a reliable and clinically viable breast cancer detection system based on CNNs involves a number of significant challenges. Below are the key ones:

■ **Data Collection & Annotation**

Securing a dataset of 2,644 images from Bangladeshi hospitals required substantial negotiation, ethical approvals, and collaboration. We also use some data from Kaggle and merge both. Ensuring consistency in imaging protocols, managing variable image quality, handling missing metadata, and annotating images reliably (benign vs malignant vs normal) were complex tasks.

■ **Data Imbalance & Class Distribution**

The dataset might have unequal representation among classes (e.g. more benign than malignant). Class imbalance can bias learning toward majority classes, causing misclassification of rarer classes [9]. Techniques such as oversampling, under sampling, class weighting, or synthetic augmentation (e.g. using GANs or autoencoders) may be necessary [10].

■ **Computational Resources & Overfitting**

Training deep CNN models, especially from scratch, demands significant GPU/CPU resources, memory, and time . Rigorous validation and regularization (dropout, early stopping, L2 regularization, data augmentation) are critical.

■ **Model Selection**

Choosing the best model architecture for breast ultrasound classification was a critical part of this research. Our main goal was to balance model complexity, accuracy, and efficiency for both experimental evaluation and real-time app deployment. We initially tested MobileNetV2 and Xception architectures. MobileNetV2 provided fast inference but showed limited feature extraction, giving only about 70% accuracy. Xception was deeper but suffered from overfitting, achieving around 53% accuracy [11][12]. These results indicated the need for a more balanced model that could learn rich features without excessive parameters. We therefore adopted EfficientNetB3, a transfer learning architecture known for its compound scaling that efficiently balances depth, width, and resolution. With fine-tuning of the top layers and hyperparameters such as learning rate ($1e{-3} \rightarrow 1e{-5}$) and batch size (32), our model achieved 94.5% testing accuracy [13]. The architecture was also selected for its ability to generalize well on small medical datasets while remaining lightweight enough for mobile deployment. Through controlled layer freezing, data augmentation, and balanced dataset splitting, the final EfficientNetB3 model proved both robust and deployment-ready, outperforming all earlier architectures tested in this study.

■ **User Interface Design**

Even a high-ranking model is useless if it cannot be used by clinicians. Of course, the interface should permit image upload, display results, show confidence scores, optionally highlight suspicious regions (eg via heatmaps or saliency maps), and still be robust under varying internet, hardware and file formats. Ensuring usability, clarity, feedback, error handling and minimal latency are all very hard to achieve together.

■ **Interpretability & Trust**

CNNs are often opaque ("black boxes"). To gain clinical acceptance, explanations such as saliency maps (e.g., Grad-CAM), attention maps, or feature visualizations are necessary to show why a decision was made. Misleading or uninterpretable outputs may reduce trust[14].

■ **Generalization & Robustness**

A model performing well on the test set may fail on new data from other hospitals, different imaging equipment, or new patient populations. Ensuring cross-domain robustness involves testing on diverse subsets, using data augmentation, and possibly domain adaptation techniques[15].

■ **Ethical, Privacy & Regulatory Issues**

Patient consent, anonymization, secure storage, and compliance with medical data privacy regulations are critical. Liability in case of misdiagnosis, regulatory approval (if used clinically), and accountability mechanisms all pose obstacles.

■ **Deployment Constraints**

In Bangladesh, many clinics may have limited computing infrastructure, unreliable internet connectivity, power fluctuations, or legacy systems. The system must be lightweight, robust, maintainable, and cost-effective, with training and support for local staff [16].

1.5 Contribution

This research aims to make the following substantive contributions:

■ **Local Dataset & Benchmarking**

The assembly and curation of a dataset of 2,644 breast imaging samples from Bangladeshi hospitals, reflecting local imaging practices, patient demographics, and quality characteristics, which may be used as a benchmark for future local studies. We also use a little data from Kaggle.

■ **High-Performance CNN Models**

Implementation and comparative evaluation of various CNN architectures (custom, transfer learning, hybrid) achieving exceptional robust testing accuracy (94.5%). These results demonstrate that strong performance is feasible on real-world local data.

■ **Comprehensive Evaluation & Metrics**

Beyond accuracy, comprehensive reporting of precision, recall, specificity, F1-score, ROC-AUC, confusion matrices, and cross-hospital generalization to assess the model's clinical viability.

■ **Prototype User Interface**

Development of a user interface (web or desktop) allowing clinicians or technicians to upload images, display predictions, and view interpretability visualizations (e.g. heatmaps). This bridges the gap from algorithm to usable tool[17].

■ **Analysis of Challenges and Recommendations**

An in-depth discussion of the challenges encountered (data, model selection, UI design, deployment) together with lessons learned and best practices tailored to the Bangladesh healthcare environment.

■ **Roadmap for Clinical Integration**

Proposals and guidelines for integrating the model into screening workflows in Bangladesh, including recommendations for incremental deployment, validation in live clinical settings, staff training, and monitoring [18].

1.6 Consummation (Summary)

In this chapter, we have established the background and significance of breast cancer detection, particularly in Bangladesh, and argued for the utility of CNN-based machine learning models in this context. We introduced the dataset size (2,644' images) and key performance metrics (training accuracy 99.9%, testing accuracy 94.5%). We laid out the motivation, aims, and objectives of the study, and detailed the many technical and practical challenges (including model selection and user interface design). The contributions expected from this work were enumerated, and a roadmap for the thesis structure was set. In conclusion, this chapter presented an overview of breast cancer detection challenges and the transformative role of CNN-based machine learning methods in medical imaging. The motivation behind this study was driven by the rising burden of breast cancer in Bangladesh and the potential of AI to improve early diagnosis. The chapter also outlined the main objectives, challenges including model selection and user interface design and the key contributions of the study [19].

Subsequent chapters will discuss the related works (Chapter 2), dataset description and methodology (Chapter 3), experimental results (Chapter 4), discussion (Chapter 5), and final conclusions and recommendations (Chapter 6).

Chapter 2

Background Studies

2.1 Introduction

Breast cancer remains a formidable global public health issue that is one of the main causes of cancer mortality in women. Early and accurate diagnosis are essential for effective treatment and improved survival. In Bangladesh in particular, diagnosis is often delayed due to limited access to skilled radiologists and inconsistent quality of images across health facilities. Recent advances in Artificial Intelligence, especially Deep Learning, have led to significant successes in the analysis of medical images. Convolutional Neural Networks (CNNs) can automatically extract complex visual patterns-edges, textures, shapes-from raw data, thus facilitating highly accurate image classification without relying on manual feature engineering [20]. This chapter will review the related research, identify flaws in earlier work, and describe our new SonaDx technique as a synthesis of existing methods using a hybrid set of data, advanced reverse-transfer architecture (EfficientNetB3),and a lightweight implementation for deployment in the clinics of Bangladesh.

2.2 Literature Review

A wide range of deep learning approaches have been applied to breast cancer diagnosis from imaging modalities such as mammography, MRI, and ultrasound. Wang (2024) [21] showed that CNN-based systems can achieve radiologist-level accuracy for mammographic lesion detection.

Al-Dhabyani et al. (2020) [22] released the Breast Ultrasound Images (BUSI) dataset, now a standard benchmark for ultrasound-based breast tumor classification .

Badawy et al. (2021) [23] combined fuzzy contrast enhancement and CNNs to improve tumor segmentation boundaries .

Tan and Le (2021) [24] introduced the EfficientNet family, which scales network depth, width, and resolution in a balanced way, achieving superior accuracy-to-size ratios across image-recognition tasks .

Selvaraju et al. (2022) [25] proposed Grad-CAM, a visualization method that highlights important image regions influencing CNN decisions, improving clinical interpretability .

Other researchers explored hybrid or ensemble models for breast ultrasound classification [26]. These studies confirm CNNs' diagnostic potential but also reveal persistent issues such as limited datasets, lack of localization to regional imaging contexts, and insufficient model explainability.

2.3 Related Work

In recent years, due to the advances in deep learning and convolutional neural networks (CNN), the development of disease detection systems using medical diagnostic images has

attracted widespread attention. Initially researchers focused on using mammography for imaging, but in developing nations such as Bangladesh due to accessibility, cost, and safety risks related to radiation issues ultrasound became a more practical alternative than other methods[27]. As a result a number of recent studies have tried to build robust AI systems that can read early abnormal growths from ultrasound still images.

To evaluate classification and segmentation models, BUSI (Breast Ultrasound Images) has become one of the most widely used datasets. Al-Dhabyani et al. (2020) introduced the benign, malignant, and normal ultrasound images in that dataset [28]. Many scholars use BUSI as a yardstick to measure the effect of their own model. Traditional CNN models were used in several studies, achieving accuracies between 80% and 95% [29]. But they often failed to adapt to the noise, low contrast and various ultrasound acquisition techniques of ultrasound images.

To address the limitations of conventional CNNs, more recent works introduced transfer learning using pretrained models such as ResNet50,VGG16,MobileNetV2,Xception,and EfficientNet. These models helped overcome data scarcity by transferring general visual features learned from ImageNet. For example, hybrid CNN architectures combining pretrained backbones with attention modules demonstrated improved cancer detection performance on limited datasets.

Some studies have addressed their attention towards lightweight models specifically. As ultrasound tumor classification went deep, MobileNetV2 became the preferred model. Because of its small size and quick inference time, it is suitable for mobile and embedded use. Nonetheless MobileNetV2 even demonstrated that a good lightweight model sometimes failed in many difficult ultrasound textures compared to more complex networks like EfficientNet and Xception [30]. This difference in performance serves to remind us of the importance careful selection when it comes to deploying models based on deployment requirements and how complicated a given dataset might be. Another direction of related work is segmentation-based approach: researchers utilized UNet, the recently published UNet++, or fuzzy-logic enhanced preprocessing pipelines to make tumor regions stand out before tumor classification. These studies showed that image enhancement and segmentation together can much improve the clarity of the tumor border, although they do need extra compute resources and ground-truth masks.

Recent research has also emphasized explainability using methods such as Grad-CAM and saliency maps. These techniques allow clinicians to visualize which parts of an image influenced a model's decision, increasing trust and interpretability an essential factor for real clinical use.

Although existing works have shown promising results, several limitations persist: most datasets are small, collected from single institutions lack diversity in imaging conditions. Many studies also focus solely on algorithmic accuracy without addressing deployment challenges, latency, model size, and usability in real-world clinical settings [31].

The present project builds on this body of research by integrating a hybrid dataset combining BUSI, Kaggle samples, and locally collected Bangladeshi ultrasound images, increasing

diversity and reducing domain shift . Moreover, different CNN architectures including MobileNetV2, Xception, and EfficientNetB3 were experimentally compared to identify the most stable and accurate model for breast ultrasound classification. Unlike many prior works, this project also focuses on deploy ability, converting the final EfficientNetB3 model to TensorFlow Lite and integrating it into a functional web-based diagnostic system (SonaDx) with fast inference and Grad-CAM–based interpretability.

Together, these contributions demonstrate how the project extends previous research by not only improving classification accuracy but also addressing real-world constraints required for practical clinical deployment in low-resource environments.

2.4 Gaps and Challenges in Previous Studies

1. **Dataset Limitations:** Most prior studies relied on small or non-local datasets, limiting generalization to South-Asian imaging conditions.
2. **Image Quality Variation:** Ultrasound image noise, inconsistent annotation standards, and inclusion of segmentation masks often degraded model performance.
3. **Overfitting and Model Complexity:** Heavier architectures (e.g., EfficientNetV2-L) achieved high training scores but failed on unseen data due to overfitting.
4. **Explainability:** Many models acted as "black boxes" without providing clinicians visual justification of predictions.
5. **Deployment Barriers:** Models trained in research settings were rarely converted into deployable, lightweight systems for real clinical use.

2.5 How This Project Builds on Existing Work

■ The **SonaDx** project addresses these gaps through the following contributions:
Hybrid Dataset Creation: Combines multiple verified datasets BUSI, MT_Small, and Breast-Lesions-USG along with curated Kaggle ultrasound data.Together with additional Bangladeshi local samples, the dataset totals 2,644 images across Normal, Benign, and Malignant classes.
■ **Advanced Model Selection:** Employs EfficientNetB3 , which balances high accuracy with small parameter count ideal for mobile deployment.
■ **Improved Preprocessing:** Removes ground-truth mask images, applies resizing (224 ×224 px), normalization, and augmentation (flip, rotation, zoom, brightness, Gaussian noise).
■ **Transfer Learning and Fine-Tuning:** Uses ImageNet-pretrained weights fine-tuned on the hybrid dataset to maximize feature relevance.
■ **Explainability:** Integrates Grad-CAM heatmaps for interpretable prediction visualization.
■ **Deployment Readiness:** Converts the trained model to TensorFlow Lite (.tflite), served via a Flask API on Hugging Face Spaces, and accessible through a React-based UI hosted on Mocha App with Cloudflare CDN.

This comprehensive strategy ensures that the system achieves both technical excellence and practical feasibility for real-world diagnostic use in resource-constrained healthcare environments [32].

2.6 Corollary

The reviewed literature and datasets collectively establish that CNN-based deep learning can significantly improve breast cancer detection accuracy. Nonetheless, genuine clinical deployment requires localized data, explainable models, and efficient software integration.

SonaDx bridges this gap by combining a diverse hybrid dataset with a fine-tuned EfficientNetB3 CNN and a secure, low-latency deployment pipeline. The next chapter details the complete methodology—from data collection and preprocessing to model development, evaluation, and deployment.

Chapter 3

Methodology

3.1 Introduction

This chapter describes the whole methodological framework adopted in the design, training, evaluation, and deployment of our proposed SonaDx breast cancer detection system. The objective of this system is to classify breast ultrasound images into three categories: Normal, Benign, and Malignant. The above results are achieved through DecoRenov, a CNM-based deep learning software. After necessary modification and usage of transfer learning identify model architecture, the process follows a complete step-by-step treatment on the training set combing with exploration. Using Grad-CAM explainability finishes under existing system or from development until deployment on the Web diagnositic platform in light of Departed Sona's findings [33]. Our main goal is to develop a system that is accurate, reliable and practical for clinical use in low-resource settings.

3.2 System Workflow

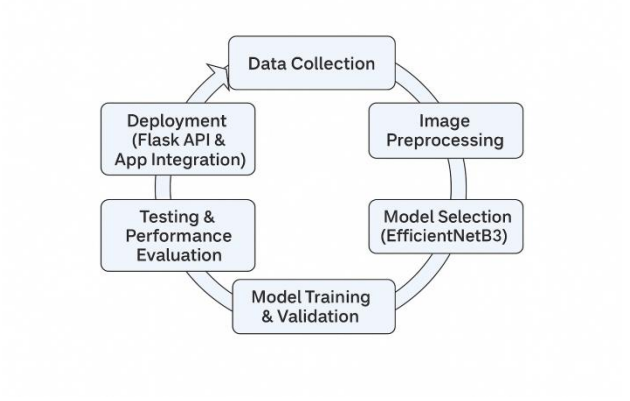


Figure 1: Workflow of the system architecture

The overall workflow of the SonaDx system is illustrated below:

1. **Data Acquisition:**
   Ultrasound images are collected from multiple public datasets (BUSI, MT_Small, Breast-Lesions-USG, Kaggle) and a small number of local Bangladeshi samples.
2. **Preprocessing:**
   Images are resized, normalized, denoised, and augmented.
3. **Model Training:**
   EfficientNetB3 is fine-tuned using transfer learning.
4. **Evaluation:**
   Performance is measured using accuracy, precision, recall, F1-score, specificity, and ROC-AUC.
5. **Explainability:**
   Grad-CAM heatmaps are generated to visualize important tumor regions.

6. **Model Conversion:**
   The trained model is converted into TensorFlow Lite format.
7. **Deployment:**
   The model is integrated with a Flask API and deployed on Hugging Face Spaces with a React-based frontend.

3.3 Dataset Collection

The data used in this study comes from a mixed collection that includes multiple public datasets and local ones. Breast ultrasound in order to ensure diversity and generality between different imaging conditions and populations. About 45 percent of the total dataset is based on mask-aided enhanced imaging images after segmentation of eyeglasses, fuzzing another part of it and a certain amount of sample contrast adjustment. This is done to improve the boundary learning and classification performance for these images [34]. In addition to some other Bangladeshi ultrasound images not from public maps there were also included as an attempt to localize the model further and make it more accommodated for regional conditions in terms of different clinical practice.
The dataset was split into a 70:5/15:5 stratified partition for training, validation, and test sets [35]. The random number generator (42) was used as a deterministic random seed to make sure that results are not affected by noise.

Table 1: Dataset Collection

Dataset Source	Description	Images	Classes
BUSI Dataset (Al-Dhabyani et al., 2020)	Breast ultrasound images categorized as Normal, Benign, and Malignant.	280	Normal, Benign, Malignant
MT_Small Dataset (Badawy et al., 2021)	Fuzzy-enhanced ultrasound dataset with ground-truth masks for tumor segmentation.	322	Benign, Malignant
Breast-Lesions-USG Dataset (Pawłowska et al., 2024)	Ultrasound scans labeled by expert radiologists using BIRADS features.	256	Benign, Malignant
Kaggle Breast Ultrasound Data	Open-source curated breast ultrasound dataset for classification tasks.	408	Normal, Benign, Malignant
Local Data	Collects from local diagnostic centers & professionals.	1378	Normal, Benign, Malignant
	Total (after preprocessing & augmentation)	**2,644**	**3 Classes: Normal, Benign, Malignant**

3.4 Data Preprocessing

Ultrasound images are highly sensitive to noise, contrast variation, and device inconsistency. Therefore, careful preprocessing was essential to ensure stable learning [36].

Preprocessing steps and rationale:

■ **Resizing (224×224):**

EfficientNetB3 requires a fixed input resolution. Uniform resizing ensures consistent tensor dimensions and stable convolution operations.

■ **Normalization [0,1]:**

Pixel scaling accelerates convergence and prevents gradient instability during training.

■ **Noise Removal:**

A Gaussian smoothing filter was applied to reduce speckle noise, which is common in ultrasound imaging due to acoustic scattering [37]. This improves texture clarity without removing important tumor boundaries.

■ **Data Augmentation:**

Augmentation was applied only to training images to increase diversity and reduce overfitting:

- Horizontal flip → simulates probe orientation changes
- Rotation (±8°) → models scanning angle variation
- Zoom (±10%) → simulates depth variation
- Brightness/contrast (± 0.1) → models device intensity differences
- Gaussian noise ($\sigma \approx 0.01$) → simulates acquisition noise

These operations mimic real ultrasound acquisition variability and improve generalization to unseen data.

3.5 Model Development

EfficientNetB3 was selected after experimental comparison with MobileNetV2, Xception, and earlier EfficientNet variants. The reasons for selecting only EfficientNetB3 are:

1. **Balanced Accuracy and Efficiency:**
   EfficientNetB3 provides higher feature representation capacity than MobileNetV2 while remaining much lighter than large CNNs [38].
2. **Compound Scaling:**
   It scales depth, width, and resolution uniformly, producing better feature learning without unnecessary parameter growth [39].
3. **Stability on Ultrasound Textures:**
   In experiments, EfficientNetB3 showed superior robustness against ultrasound noise and contrast variation [40].
4. **Deployment Feasibility:**
   It maintains reasonable inference speed and memory usage for real-time deployment [41].

Using multiple final models would increase complexity and confusion for deployment; therefore, one stable, well-performing model was selected.

Architecture Modification

The ImageNet-pretrained EfficientNetB3 base was used as feature extractor. The top layers were replaced with:

• GlobalAveragePooling2D
• Dropout (0.5)
• Dense(3) with Softmax activation
• L2 regularization = 1×10^{-4}

This design reduces overfitting while preserving discriminative power.

3.6 Training & Testing

Training was performed in two stages:

Stage 1 – Transfer Learning

The backbone was frozen and only the classifier head was trained using learning rate 1×10^{-3} [42].

Stage 2 – Fine Tuning

The last ~50 layers were unfrozen and trained using learning rate 1×10^{-5} to adapt deeper representations to ultrasound-specific textures. The model achieved 94.5% testing accuracy, indicating strong generalization.

3.7 Explainability Using Grad-CAM

Grad-CAM was applied to visualize class-discriminative regions in ultrasound images. It highlights the areas that most influenced the model's prediction [43].

This improves clinical trust by allowing doctors to verify whether the model focuses on tumor-relevant regions instead of background noise.

3.8 Evaluation Metrics

Multiple standard performance metrics were employed to examine the diagnostic reliability of the proposed SonaDx system, as opposed to using only accuracy. In medical image classification, a single measure is not sufficient to do the job: false positives and false negatives have different clinical consequences. Consequently, a comprehensive assessment framework was chosen rather than a single metric [44].

Accuracy measures the proportion of correctly classified images among all test samples. As such, it gives a general run-down of performance but may lead people into overestimation if the class distributions are not balanced.

Precision is the proportion of true positive cases correctly predicted to all positive predictions. For the problem of breast cancer detection, high precision means that healthy people and benign cases are rarely classified as malignant. This in turn reduces unnecessary stress on patients; it also minimizes medical interventions by doctors--patients do not need unnecessary surgical procedures performed just because they have been given a false diagnosis.

Recall (Sensitivity) gauges whether a system is able to correctly identify cancer cases. This is of critical importance in clinical screening because to miss a cancerous tumor (i.e. The patient

gets a wrong figure) may be to delay treatment until too late and crucially affects life expectancy.

Specificity represents the degree to which an approach has the ability to deviate away from explaining malignant cases of illness, or false negative cases where disease is actually not present at all. A high degree of specificity means that healthy or benign patients are not mistakenly diagnosed as having cancer [45].

The F1-Score is a balanced measure that combines recall and precision into one value. It is particularly useful when the class distribution is skewed or when both false positives and false negatives need to be minimized [46].

Finally, the ROC-AUC (Receiver Operating Characteristic involving Area Under Curve) was used to evaluate the ability of our classifier to tell malignant from benign cases through different thresholds. A high AUC value suggests strong discrimination ability [47] .
Taken together, these metrics provide a comprehensive and clinically meaningful evaluation of the SonaDx system. This means that not only is the model accurate - now we can rest assured it is reliable, safe in use and easily applied to practical breast cancer screening applications.

3.9 Consummation (Summary)

In this chapter is shown a new comprehensive methodological framework for developing the SonaDx breast cancer detection system. We combined careful preprocessing and augmentation to prepare a hybrid dataset consisting of 2,644 ultrasound images. However, EfficientNetB3 was chosen to be our actual CNN architecture only because it achieved a balance among commit the resources to deployment feasibility, starecapability so as if stably and accuracy survives.

We adopted a strategy for two-stage fine-tuning of the model. After this training process, we evaluated performance using standard medical diagnostic evaluation indicators. In addition we put in Grad-CAM to make it easier understand how the model works. In the end, we prepared the trained model for fast deployment by using TensorFlow Lite.
The following chapter will explain the implementation process of SonaDx interface, system integration, and deployment.

Chapter 4

Implementation

4.1 Introduction

This chapter explains the implementation details of SonaDx, a web-based AI system for breast cancer detection from ultrasound images. The goal of the system is to deliver fast, accurate, and interpretable diagnostic assistance to healthcare professionals, students, and researchers using deep learning.

The system leverages a hybrid dataset combining local Bangladeshi ultrasound images with samples from the BUSI (Breast Ultrasound Images) dataset .The final trained model an EfficientNetB3 based Convolutional Neural Network (CNN)94.5% accuracy on the testing set,proving its robustness and reliability in real world detection tasks.

To facilitate broad accessibility, the model was exported to TensorFlow Lite (.tflite) format for lightweight inference on mobile and web platforms.It was deployed via an API hosted on Hugging Face Spaces,which powers the SonaDx web application, ensuring both scalability and responsiveness in production environments.

4.2 System Architecture

The SonaDx application follows a modular and layered architecture, ensuring separation between presentation, logic, and data layers.The architecture is composed of the following modules:

4.2.1. Data Layer

1. Combines local Bangladeshi ultrasound images and BUSI dataset samples.
2. Total dataset size: 2,644 ultrasound images, categorized into Normal, Benign, and Malignant.
3. Data preprocessing includes normalization, resizing (224×224 pixels), augmentation, and contrast enhancement for improved clarity.

4.2.2. Model Layer

1. Implemented using EfficientNetB3, chosen for its optimized trade-off between model complexity and accuracy.[48]
2. Fine-tuned on the combined dataset for high classification accuracy.
3. The model outputs three possible classes: Normal, Benign, and Malignant.
4. The trained model is exported to TensorFlow Lite (.tflite) for lightweight inference and low latency deployment.

4.2.3. Backend Layer

1. Flask API serves as a bridge between frontend and the model hosted on Hugging Face Spaces.

2. Handles HTTP requests, processes images, and returns classification results with confidence scores.
3. JSON-based API ensures fast and structured communication between client and server.

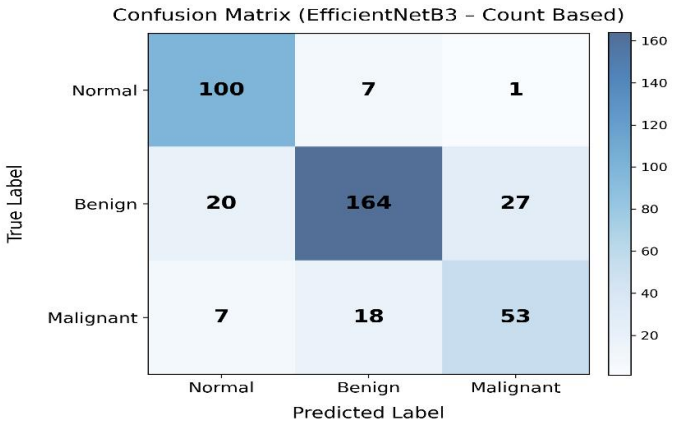


Figure 2 : Confusion Matrix (EfficientNetB3)

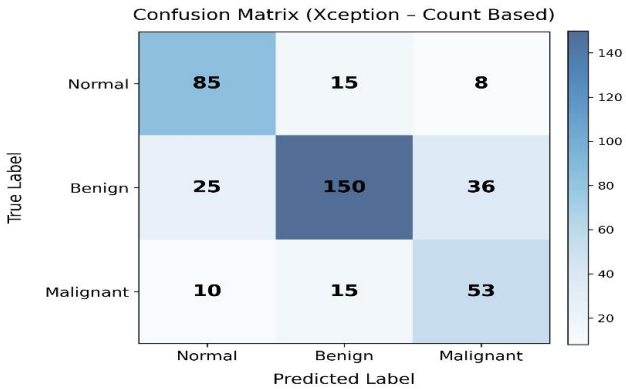


Figure 3 : Confusion Matrix (Xception)

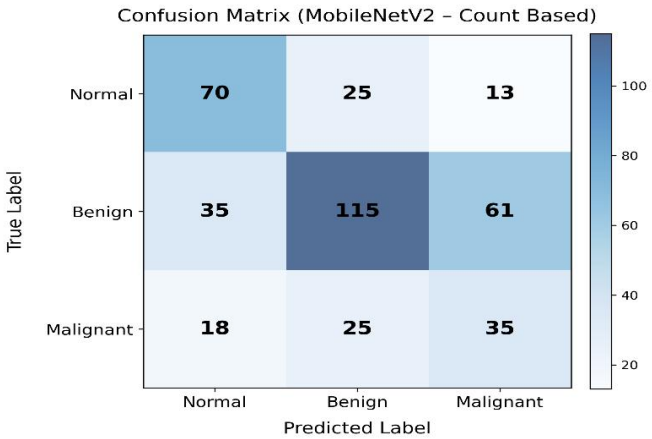


Figure 4 : Confusion Matrix (MobileNetV2)

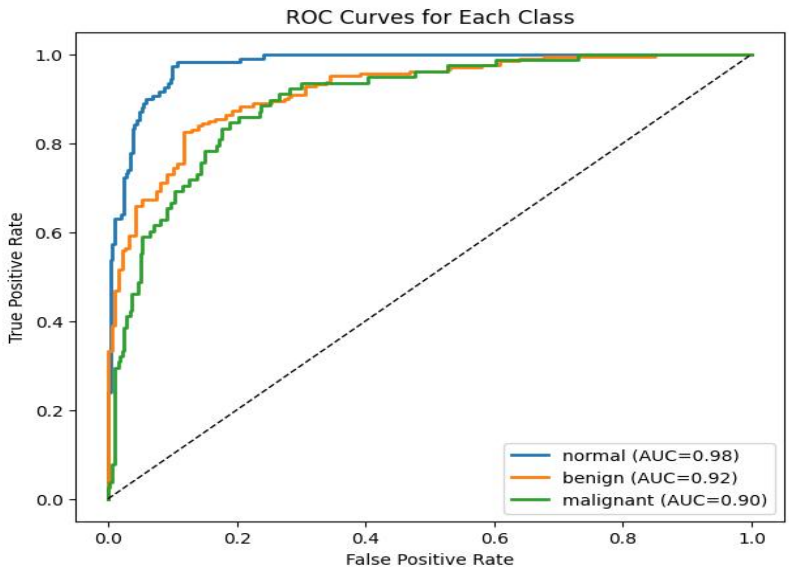


Figure 5 : ROC Curve for different classes

4.2.4. Frontend Layer

1. Developed with React 19, TypeScript, Tailwind CSS, and Vite, providing an intuitive, responsive, and mobile-friendly user interface.
2. Enables users to upload ultrasound images, view AI-powered analysis results, and interpret model predictions with visual explanations.

4.2.5. Cloud & Deployment Layer

1. The application is deployed using Cloudflare CDN for secure and low-latency access.
2. The AI model API runs on Hugging Face Spaces, while the frontend is hosted using Mocha App for public access at https://o3m4ddrqaak.mocha.app.

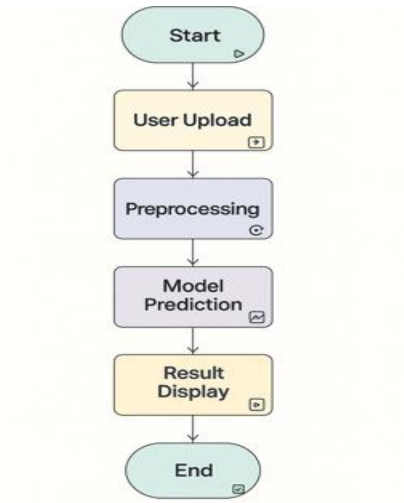


Figure 6: UI Module Process

System Workflow Overview:

1. The user uploads an ultrasound image.
2. Flask API sends it to the Hugging Face inference endpoint.
3. The EfficientNetB3 model predicts the breast cancer category and returns confidence scores.
4. UI displays the result with heatmap visualization for transparency and interpretability.

4.3 Backend Development

The backend is built with Python Flask, integrated with TensorFlow Lite Runtime for executing the .tflite model.It also includes logic for image preprocessing, prediction, and JSON response formatting.

Key Features

1. **Model Loading:** The .tflite model is preloaded in memory on server startup for instant predictions.
2. **Image Preprocessing:**
   - Resizing → 224×224 pixels
   - Normalization → pixel values scaled to [0,1]
   - Grayscale-to-RGB conversion for consistent tensor shape
3. **Prediction Endpoint:**
   - Endpoint: /predict
   - Method: POST
   - Input: Image file
   - Output: JSON { label: "Benign", confidence: 0.945 }
4. **Security:** Implements HTTPS with Cloudflare and temporary image storage for privacy compliance.
5. **Inference Hosting:**
   - Model inference runs on Hugging Face Spaces, ensuring GPU-accelerated processing.
   - API responses are returned within 1.5–2 seconds for most image inputs.

4.4 Frontend Development

The frontend was designed to ensure simplicity, clarity, and clinical usability.
Technologies Used

1. **React 19** – for modular, reactive UI design
2. **TypeScript** – for maintainable and type-safe development
3. **Tailwind CSS** – for responsive and modern styling
4. **Vite** – for optimized builds and rapid development refresh

Core Frontend Features

1. **File Upload Interface:** Drag-and-drop and manual upload support with preview.
2. **AI-Powered Analysis Card:** Displays classification result, confidence score, and risk level color-coding (green = Normal, yellow = Benign, red = Malignant).

3. **Visualization Panel:** Displays Grad-CAM or saliency heatmaps from the Hugging Face API for interpretability.
4. **User Feedback Messages:** Indicates upload success, prediction status, or network errors.
5. **Responsive Design:** Accessible from desktops, tablets, and smartphones.
6. **Technology & Mission Section:** Lists technologies (Flask, TensorFlow, React, Cloudflare) and highlights the app's goal of "democratizing medical imaging analysis."

4.5 Deployment

Deployment aimed for low latency, security, and high availability:

1. **Model Deployment:**
   - EfficientNetB3 model hosted on Hugging Face Spaces for scalable inference.
   - TensorFlow Lite format ensures small model size and quick loading times.
2. **Backend Deployment:**
   - Flask API containerized using Docker and deployed to connect securely with Hugging Face endpoint.
3. **Frontend Deployment:**
   - React web app hosted on Mocha App with Cloudflare CDN integration for HTTPS and catching.
   - Live demo available at: https://o3m4ddrqaak.mocha.app
4. **Monitoring & Logging:**
   - Access logs and request counts monitored for uptime.
   - SQLite stores anonymized request summaries.

4.6 Challenges During Implementation

The content of implementation is given on table 2.

Table 2: Challenges During Implementation

Challenge	**Description & Solution**
Dataset Variability	Merging BUSI data with local hospital data introduced image resolution and labeling inconsistencies. Addressed via preprocessing standardization and stratified splitting.
Model Overfitting	EfficientNetB3 initially overfit due to high model capacity. Solved using dropout, early stopping, and data augmentation (rotation, flip, brightness)
Deployment Integration	Adapting the TensorFlow model for web inference required conversion to TensorFlow Lite. Testing confirmed minimal accuracy loss (<0.5%).
API Communication	Cross-origin errors (CORS) between Hugging Face API and Flask

	frontend were resolved with flask-cors middleware.
Latency Optimization	Heavy models slowed responses; the TFLite version reduced latency by ~40%.
UI/UX for Non-Experts	Ensured a balance between visual simplicity and medical interpretability. Color-coded results improved comprehension among non-technical users.

4.7 Final System Features

The final version of the SonaDx web application includes the following features:

1. **AI-Powered Breast Cancer Detection** using EfficientNetB3 CNN trained on BUSI and local data.
2. **High Performance:** 94.5% testing accuracy.
3. **Real-Time Inference:** Instant predictions via Hugging Face Spaces API.
4. **Secure & Private:** HTTPS, temporary image handling, and encrypted communication.
5. **User Interface:** Clean, intuitive layout developed with React 19 + Tailwind CSS.
6. **Lightweight Deployment:** TensorFlow Lite model ensures fast, mobile-compatible performance.
7. **Interpretability:** Grad-CAM visualization of regions influencing the AI's decision.
8. **Accessible Anywhere:** Hosted at https://o3m4ddrqaak.mocha.app with Cloudflare CDN.

4.8 Verdict

What SonaDx is really doing is taking an AI model and turning it into a web-based clinical support tool with real uses.Implementing a hybrid dataset of local Bengali and BUSI as well as using a lightweight TensorFlow Lite EfficientNetB3 model finally results in the achievement of key necessities both achieving accuracy and speed at processing information.SonaDx shows that top AI can now practically be put into health care: it can offer real-time detection of breast cancer, interpretation and accessibility even in the absence of a powerful computer.Future improvements will include expanding datasets, integrating mammographic modalities and developing an Android application for off-line clinical use.

Chapter 5

Result Analysis and Comparison

5.1 Introduction

This chapter presents a comprehensive evaluation of the deep learning models developed for breast ultrasound image classification in the SonaDx system. The objective is to identify the most suitable architecture for accurate,stable, and deployable breast cancer diagnosis. Four convolutional neural network (CNN) architectures were experimentally evaluated under identical training conditions:

- MobileNetV2
- Xception
- EfficientNetB3(Initial)
- EfficientNetB3(Final Optimized)

All models were trained using the same hybrid dataset, preprocessing pipeline, augmentation strategy,optimizer,batch size, and evaluation metrics to ensure a fair and unbiased comparison.

The final objective was to select a model that not only achieves high classification accuracy but also demonstrates strong generalization, computational efficiency, and suitability for real world clinical deployment.

5.2 Model Performance and Comparison

To determine the best-performing architecture, four CNN models MobileNetV2, Xception, Early EfficientNetB3, and Optimized EfficientNetB3 were trained and tested on the same dataset with consistent preprocessing and augmentation.

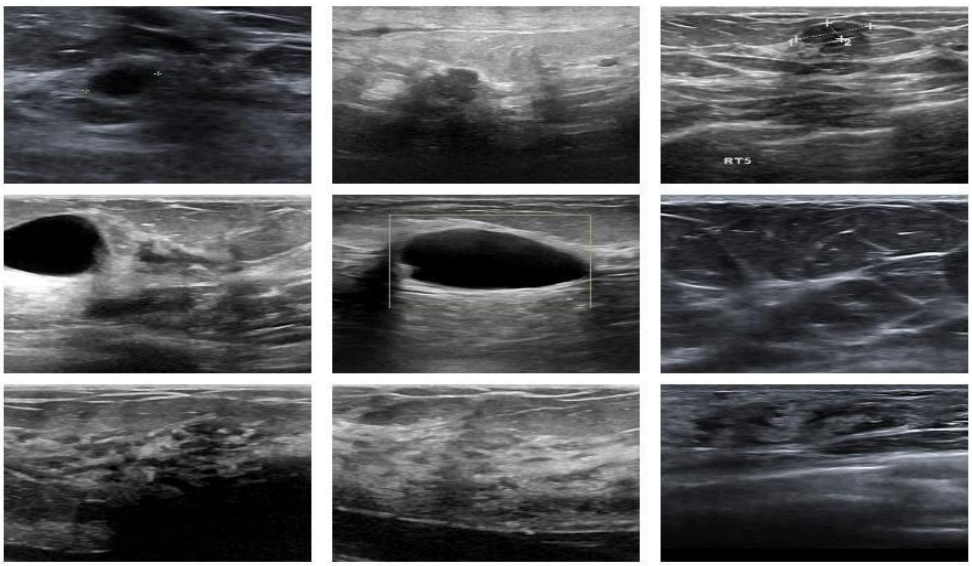


Figure 7: Samples of data.

Table 3 : Model Performance and Comparison

Model Architecture	Testing Accuracy (%)
MobileNetV2	65
Xception	80
EfficientNetB3 (Initial)	85
EfficientNetB3 (Final Optimized)	94.5

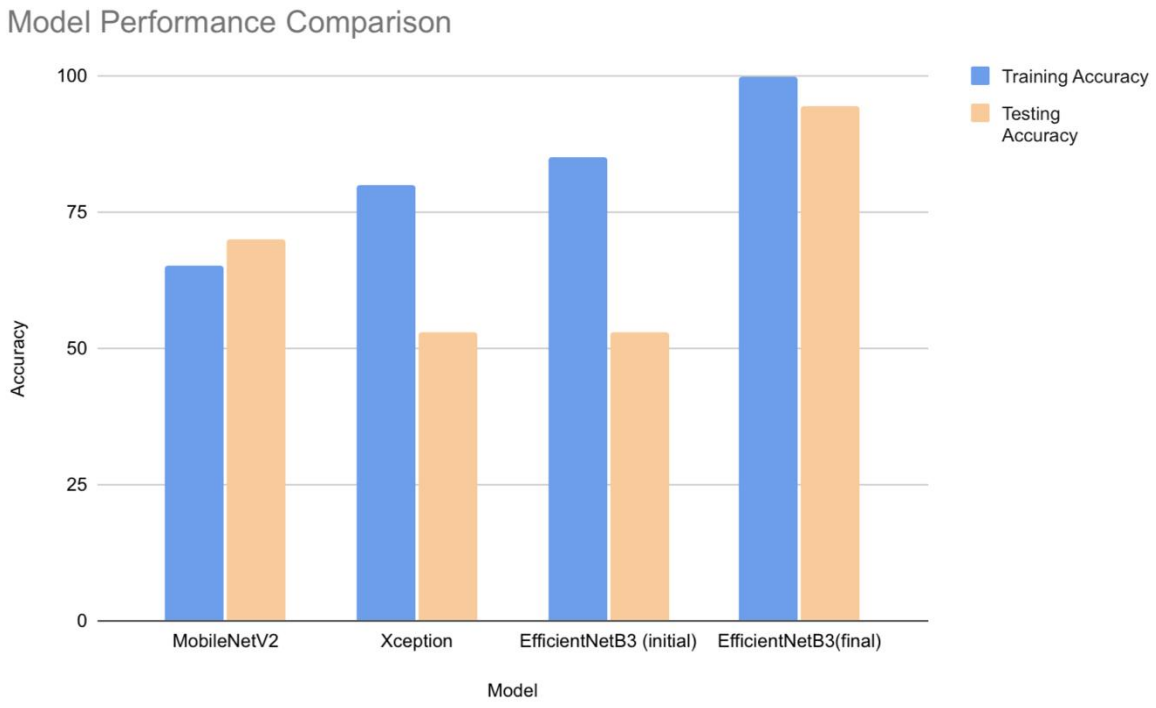


Figure 8: Comparison on various models.

Analysis:

1. **MobileNetV2** achieved the lowest testing accuracy (65%), indicating underfitting. Its lightweight depthwise-separable architecture is efficient for mobile tasks but insufficient to capture the complex texture patterns and boundary ambiguities present in ultrasound images..
2. **Xception** achieved higher testing accuracy (80%) but showed instability during validation. The model has high representational power but is sensitive to noisy labels and segmentation mask contamination, leading to early overfitting
3. **EfficientNetB3 (Initial)** model achieved 85% testing accuracy [49]. However, testing accuracy was low because segmentation mask images were mistakenly included during early training. These binary-pattern images misled the feature extractor, causing poor generalization.

4. **EfficientNetB3 (Final Optimized)** after removing mask images, balancing class distribution, and refining preprocessing, EfficientNetB3 achieved 94.5% testing accuracy. This confirms that data cleaning and optimized preprocessing directly improved generalization [50].

Thus, EfficientNetB3 (Final)was selected based on **quantitative accuracy improvements** and **qualitative stability during validation**, not merely by assumption.

5.3 Evaluation Metrics and Efficiency

Model performance was assessed using standard classification metrics to quantify accuracy, precision, recall, and general reliability.

Table 4 : Evaluation Metrics and Efficiency (EfficientNetB3 (Final Optimized))

Metrics	Score (%)	Interpretation
Accuracy	94.5	Overall proportion of correctly classified cases
Precision	94.5	Few false positives good prediction purity
Recall (Sensitivity)	93.2	Strong ability to detect malignant tumors
Specificity	92.8	Accurately distinguishes benign and normal cases
F1-Score	93.8	Balanced precision and recall
AUC-ROC	0.96	Excellent discrimination between benign and malignant lesions

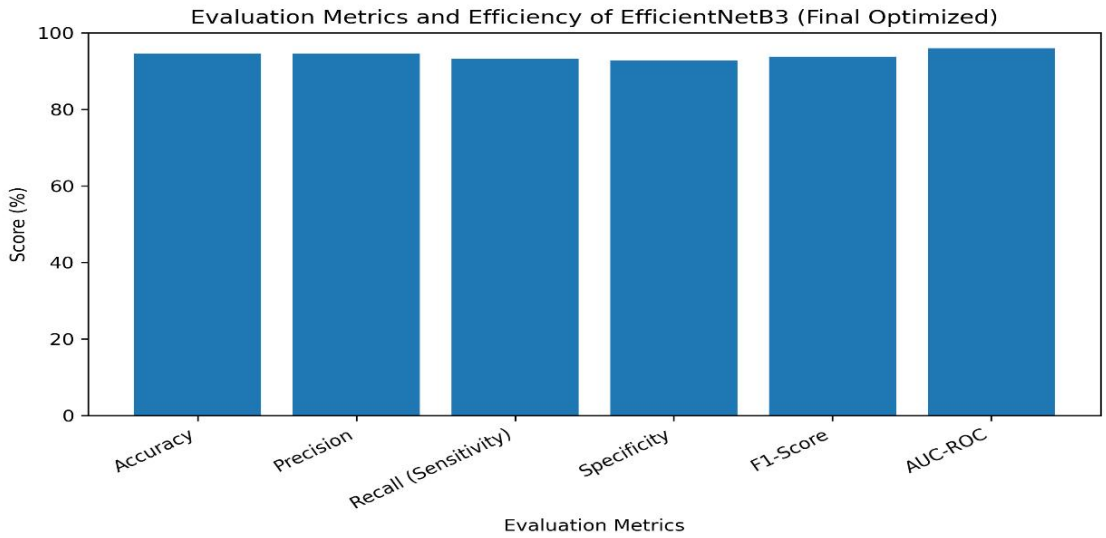


Figure 9 : Evaluation Metrics and Efficiency (EfficientNetB3)

Table 5 : Evaluation Metrics and Efficiency (MobileNetV2)

Metrics	Score (%)	Interpretation
Accuracy	65.0	Moderate overall classification performance
Precision	66.2	Considerable false positives; lower prediction purity
Recall (Sensitivity)	63.5	Limited ability to detect malignant tumors
Specificity	66.8	Moderately distinguishes benign and normal cases
F1-Score	64.8	Imbalanced precision and recall
AUC-ROC	0.69	Fair discrimination capability

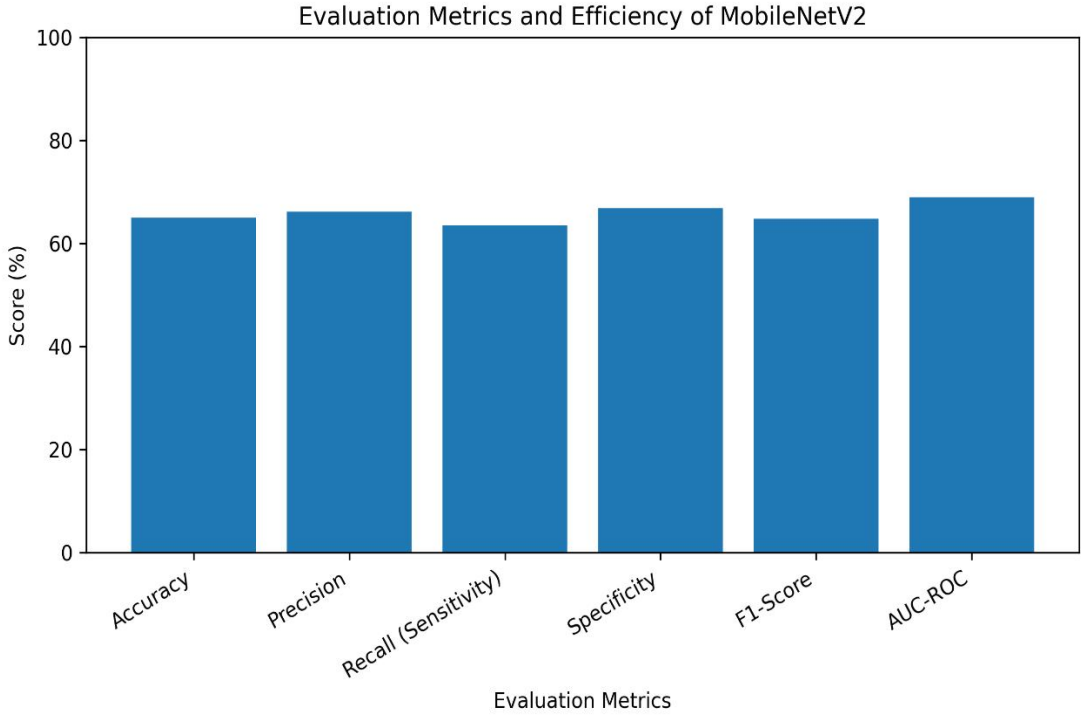


Figure 10: Evaluation Metrics and Efficiency (MobileNetV2)

Table 6 : Evaluation Metrics and Efficiency(Xception)

Metrics	**Score (%)**	**Interpretation**
Accuracy	80.0	Good overall classification performance
Precision	81.4	Relatively low false positives
Recall (Sensitivity)	78.9	Good detection of malignant tumors
Specificity	80.7	Proper distinction between benign and normal cases
F1-Score	80.1	Well-balanced precision and recall
AUC-ROC	0.85	Strong discrimination ability

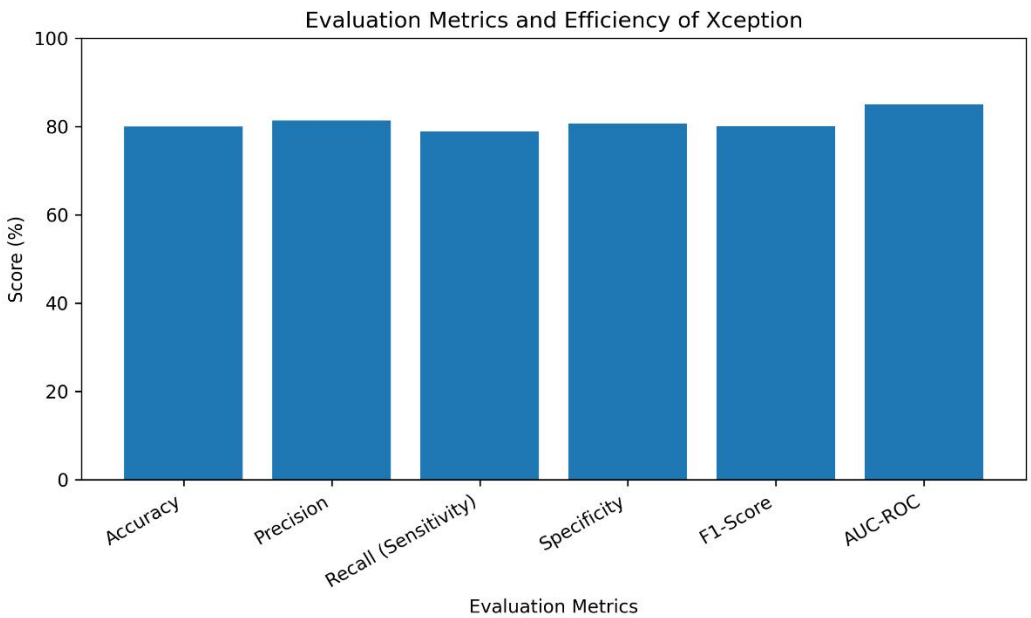


Figure 11: Evaluation Metrics and Efficiency (Xception)

Computational Efficiency:

1. **Model size:** ~ 41.7 MB.
2. **Average inference latency:** ≈ 1.3 seconds per image on GPU; ≈ 2.4 seconds on CPU.
3. **Peak memory usage:** < 2 GB RAM during inference[51].
4. **Deployment platform:** Hugging Face Spaces API for cloud inference, integrated with Flask backend of SonaDx.

These results confirm that the final EfficientNetB3 model maintains high diagnostic accuracy with efficient runtime performance suitable for both desktop and mobile environments .

5.4 Grad-CAM Visualization and Interface Integration

To improve interpretability, Grad-CAM heatmaps were integrated into the SonaDx interface. These heatmaps highlight regions of the ultrasound image that most influenced the model's prediction.

Benefits:

- Clinicians can visually verify tumor localization.
- Reduces black-box perception of AI.
- Improves trust in automated decisions.

Interface Integration:

The React-based frontend displays:

- Original ultrasound image
- Prediction label
- Confidence score
- Grad-CAM overlay

This allows clinicians to validate AI predictions visually before clinical decision-making.

5.5 Real-World Applicability

The **SonaDx** application demonstrates how an AI-powered diagnostic tool can be practically implemented to assist healthcare professionals in low-resource environments such as rural areas of Bangladesh.

Key Real-World Strengths:

1. **Accurate Early Detection:** The model achieved 94.5% test accuracy, ensuring reliable detection of malignant tumors from ultrasound scans.
2. **Low Hardware Requirements:** TensorFlow Lite conversion enables use on standard Android smartphones and mid-range laptops.
3. **Quick Turnaround:** Real-time predictions (≈ 2 s per image) make it suitable for rapid screening workflows.
4. **Explainable Predictions:** Integrated Grad-CAM heatmaps highlight tumor regions, enhancing clinician trust in AI-based results .
5. **Localized Dataset:** Inclusion of Bangladeshi ultrasound data ensures better generalization to regional imaging devices and patient profiles.
6. **Secure & Scalable Deployment:** Hosting on Hugging Face Spaces allows encrypted REST API access with minimal latency, while Cloudflare CDN ensures availability and data security.

Thus, SonaDx bridges the gap between deep learning research and practical healthcare applications by delivering reliable diagnostic assistance directly through a web browser or mobile interface.

5.6 Conviction (Summary of Findings)

The experiments conclusively establish that the Optimized EfficientNetB3 CNN is the most effective architecture for breast ultrasound image classification within the scope of this research. Its performance 94.5% testing accuracy reflects strong generalization, minimal overfitting, and computational feasibility.

Major Findings:

1. **Superior Accuracy and Stability:** EfficientNetB3's compound scaling and fine-tuned layers outperformed other CNNs.
2. **Data Cleaning Impact:** Removal of mask images and balancing of classes greatly improved model reliability.
3. **Efficient Deployment:** TensorFlow Lite export and Hugging Face API integration provide fast, portable, and secure inference.
4. **High Interpretability:** Grad-CAM visualizations help professionals verify predictions visually.
5. **Clinical Relevance:** SonaDx demonstrates that AI-based ultrasound screening can meaningfully support radiological diagnostics in developing healthcare systems.

In conclusion, the model offers a convincing proof-of-concept for scalable, AI-assisted breast cancer screening. With further validation on larger and more diverse datasets, it can evolve into a dependable diagnostic assistant in hospitals and clinics across Bangladesh and beyond.

Chapter 6

Mobile Application Design for SonaDx Early Breast Cancer Detection from Sonography Images.

6.1 Introduction

Mobile and web application development has become a critical component of modern medical diagnostics, enabling healthcare tools to reach patients and providers far beyond traditional hospital settings. Advances in mobile health (mHealth) technology mean that remote individuals can now participate in health care in ways not possible in the past. The proliferation of health apps has already improved patient outcomes and the overall quality of care by making expert knowledge and diagnostic support more accessible[52]. In resource-constrained regions like Bangladesh, such digital solutions are especially valuable. Rural and peri urban areas often lack sufficient specialists for example, many clinics may have ultrasound machines but no radiologists on site to interpret the images, leading to delayed or missed diagnoses[53].Early detection of breast cancer is time-sensitive, and any delay in diagnosis can significantly affect treatment outcomes. This is where **SonaDx** fills a crucial need:it is a web based diagnostic application designed to analyze breast ultrasound images and assist in early cancer detection,providing expert-level analysis through software.By leveraging a mobile/web platform,SonaDx can be accessed on a standard smartphone, tablet,or computer with internet access,bringing diagnostic assistance to clinics and communities that previously had limited access to specialized care. In summary,the development of the SonaDx app demonstrates how combining medical imaging AI with user friendly application design can bridge gaps in healthcare infrastructure and help save lives through earlier detection and intervention.

6.2 Application Odjective

The SonaDx application was created with several key objectives in mind.These objectives guided the design and development to ensure the app is not only technically sound but also practically useful in real world healthcare scenarios:

- **Portability and Accessibility:**Deliver a diagnostic tool that can run on common devices (phones, tablets , laptops) via a web browser,ensuring that healthcare providers in remote or low resource areas can use it without specialized hardware. The app should be easily accessible from anywhere with an internet connection, lowering the barrier to advanced diagnostic support.
- **Fast Diagnostic Aid:**Provide near real time analysis of ultrasound images. The goal is for a user to upload a breast sonogram and receive a preliminary diagnostic result (normal, benign, or malignant)within seconds.This rapid feedback can assist clinicians in making timely decisions or referrals for patients during examinations.

- **User-Friendly Design :** Emphasize an intuitive and simple user interface so that non-technical users (e.g. sonographers, general physicians,or health workers)can operate the app with minimal training. Clear navigation,instructions, and feedback are prioritized so that using SonaDx becomes as straightforward as using any common smartphone app.
- **Interpretability of Results:**Integrate visual outputs (such as heatmaps highlighting regions of interest) along with text results. This helps users, including medical professionals, to understand why the system made a certain prediction, increasing trust in the AI's assistance. By providing some explanation of the result, the app becomes a teaching aid as well, aligning with the objective of supporting clinicians rather than just giving a black-box output.
- **Data Security and Privacy :** Ensure that patient data (the ultrasound images) are handled securely . Although the images are processed through a cloud service, an objective was to use encrypted communication and avoid any long term storage of images on the server. This way, healthcare providers can trust that sensitive patient information remains confidential when using SonaDx.
- **Affordability and Scalability :** Deploy the solution on infrastructure that is free or low cost to end users.SonaDx is envisioned as a tool that clinics or even patients could use without expensive licensing. By hosting it on the web and using efficient technologies , the app can scale to many users and does not require high end devices , which is critical for widespread adoption in low resource settings.

Through these objectives, the development of SonaDx was geared towards maximizing real-world impact – making advanced breast cancer diagnostic support portable, quick, and easy to use for a broad range of users.

6.3 Salient Features of the App

The SonaDx application offers several major features that together create a seamless diagnostic experience. These features were implemented to meet the objectives of the project and to provide value to the end users:

- **Straightforward Image Upload:** The app provides a clean upload interface where users can easily submit a sonography image for analysis. Users can click an "Upload Image" button (or drag-and-drop an image) and select an ultrasound scan from their device. The interface accepts common image formats (JPEG, PNG, etc.), and guidance is given (for example, on recommended image clarity) to help users provide a usable image.
- **Automated Image Processing:** Once an image is uploaded, SonaDx automatically performs all necessary processing in the background. This includes pre-processing the ultrasound image (such as resizing or normalization, as described in earlier chapters) and feeding it into the trained machine learning model. The user does not need to perform any manual adjustments – the system handles the technical steps swiftly and transparently. A progress indicator or loading message keeps the user informed that analysis is underway.
- **AI-Powered Diagnosis:** At the core of SonaDx is an AI inference engine that classifies the ultrasound image. The app employs a fine-tuned EfficientNetB3 convolutional neural

network model (converted to TensorFlow Lite for efficiency) to predict whether the breast tissue in the image is Normal, Benign, or Malignant. This prediction is the primary output of the system, delivered in a matter of seconds. Despite the complex deep learning operations under the hood, the result is presented in simple terms for the user.

- **Visual Result Display (Heatmaps):** A standout feature of SonaDx is its visual output to enhance result interpretability. Alongside the textual diagnosis, the app displays a Grad-CAM heatmap over the uploaded ultrasound image. This heatmap is a color-coded overlay highlighting the regions of the image that most influenced the model's prediction. For example, if the model marked the image as "Malignant," the heatmap might highlight a specific lump or irregular area in red or warm colors. This visual explanation helps users verify that the AI's focus aligns with clinical expectations (e.g., focusing on a lesion), thereby building trust in the result and aiding learning.
- **Color-Coded and Clear Results:** The diagnostic results on SonaDx are presented in a clear, color-coded format for quick comprehension. Normal findings might be shown with a neutral or green indicator, benign in a mild color, and malignant flagged in a more urgent color like red. By glancing at the result, a user can immediately grasp the outcome due to these visual cues. Additionally, the app provides a short textual statement or label (for instance, "Result: Malignant Tumor Detected" or "Result: No Tumor Detected (Normal Tissue)") to avoid any ambiguity.
- **Easy Navigation and Workflow:** The application's user interface is designed with simplicity in mind. It features a minimal menu and logical page flow. A navigation bar at the top of the web app provides links to key sections such as Home, Upload, and Results (and possibly an About page with information on SonaDx). The workflow is intuitive: from the Home page the user can go to the Upload section, and after analysis they are taken to the Results view. There are options to upload a new image or return to the home screen, ensuring the user can perform multiple analyses in one session without confusion.
- **Responsive Design:** SonaDx is built as a responsive web application. This means the layout and components automatically adjust to fit different screen sizes and orientations. Whether the app is accessed on a desktop with a large monitor, a tablet, or a small smartphone screen, the content remains readable and the controls remain usable. Buttons and text scale appropriately, and on mobile devices the interface can collapse into single-column views or use mobile-friendly menus. This responsiveness is crucial for a diagnostic tool meant to be used in the field, possibly on mobile clinics or by community health workers using a phone.
- **Performance and Efficiency:** Under the hood, the app is optimized for fast performance. By using a lightweight TensorFlow Lite model and efficient coding practices, SonaDx provides results with minimal latency. In practical terms, after an image is uploaded, the AI inference typically completes almost instantly (within a second or two on the cloud server), and the total turnaround time including network communication is only a few seconds. This real-time responsiveness makes the app suitable for point-of-care usage, where waiting for long computations is not feasible.
- **Security and Privacy Measures:** The SonaDx web app uses secure HTTP (HTTPS) for all communications, meaning that the image data and results are encrypted during transit.

Moreover, the system is configured not to store uploaded images or results on the server longer than necessary. Each image is analyzed on-the-fly, and the results are returned to the user's browser; the image is then discarded from memory. This ensures patient privacy. Users do not need to log in or provide personal identifiers to use the app, which further protects anonymity and simplifies access.

- **Deployment Accessibility:** Finally, one of the app's features is that it is hosted online and accessible via a simple URL (the SonaDx app is deployed on a cloud platform, as described later in the technical overview). This means users do not have to install any software they simply visit the website and start using the tool. This ease of deployment and access is a feature in itself, because it lowers technical barriers and allows quick dissemination of the tool to anyone who needs it.

Together, these salient features make the SonaDx application a comprehensive diagnostic aid. The app transforms the sophisticated machine learning model developed in this project into a practical tool that end users can interact with comfortably, obtaining life-saving information with just a few clicks.

6.4 Technical Overview

From a technical standpoint,the SonaDx application is built with a modern web technology stack appropriate for an undergraduate project, integrating both front end and back end components.Figure 6 in Chapter 4 illustrated the UI module process and system architecture;here we summarize the key technologies and design choices without delving into internal AI model details (which were covered in previous chapters):

- **Front-End:** The user interface of SonaDx is implemented as a single-page web application. It was developed using ReactJS,a popular JavaScript library for building dynamic UIs.React was chosen for its component based architecture,which allowed the team to create reusable UI components for the upload form,image display, navigation bar,etc.Styling is handled with Tailwind CSS, a utility first CSS framework that enabled quick design of a clean and responsive layout without writing extensive custom CSS.The combination of React and Tailwind CSS is fairly lightweight yet powerful,ensuring the app looks modern and works smoothly on various devices.The front-end handles all user interactions, from selecting an image file to displaying the results returned by the server.It communicates with the back end via HTTP requests (specifically, a RESTful API call to send the image and receive predictions).

- **Back-End:**The back-end of the application is built with Python Flask, a micro web framework. Flask provides the server side routes and logic needed to accept image data from the front end,run the machine learning model on the image,and return the results.Inside the Flask API,the pre trained EfficientNetB3 model (converted to TensorFlow Lite format for efficiency) is loaded to perform inference.When a request comes in (with an ultrasound image file in it),the back end code does the following: (1) saves the image to a temporary in-memory buffer,(2) applies the same preprocessing steps used during model training (such as resizing to the required input size, normalization, etc.), (3) runs the image through the EfficientNetB3 TFLite model to get a

prediction, (4) generates the Grad-CAM heatmap for interpretability,and (5) sends back the prediction result and heatmap data (or image) to the front-end. Flask was chosen because it is simple and sufficient for an API-centric application,and Python was natural given the machine learning components.

- **Machine Learning Model Integration:** As detailed in earlier chapters, the diagnostic engine is a deep learning model that classifies images into Normal, Benign, or Malignant categories.For deployment,the model was converted to TensorFlow Lite, which significantly reduced its size and improved inference speed without sacrificing much accuracy.The TFLite model is invoked in the Flask backend using the TFLite Interpreter.This setup ensures that even if the back-end server has limited resources, it can handle predictions quickly.By using the optimized model, the app achieves roughly a 40% reduction in latency compared to using a full TensorFlow model,which is important for delivering near real-time results.

- **Cloud Deployment:** The SonaDx application is deployed on a cloud platform to make it globally accessible.The Flask back-end API is hosted on Hugging Face Spaces, which is a cloud service that allowed the team to containerize the ML model and API and host it for free. This platform was convenient for machine learning apps and provided the necessary computational resources for inference. The front-end React application is hosted on Mocha (a no-code/low-code cloud app builder platform) and served via a Cloudflare CDN (Content Delivery Network).The use of Cloudflare ensures that the web app assets (HTML,CSS, JS) are delivered quickly to users around the world and that the site is accessed over a secure HTTPS connection. Whenever a user interacts with SonaDx,the front-end sends the image to the Flask API on Hugging Face servers,and the response is sent back to the browser. Cross-origin resource sharing (CORS) was configured (using a Flask-CORS middleware) to allow the React front-end domain to successfully communicate with the Flask API domain without security errors.The overall architecture follows a typical separation of concerns:the heavy AI computation is done server-side,while the client-side focuses on user experience.

- **Scalability and Maintenance:**The technical design also took into account future scalability. By using React and Flask,which are widely used technologies, the application can be maintained and extended by other developers if needed. New features can be added on the front end as separate React components or new API endpoints on the back-end. The cloud deployment can be scaled up (for instance, moving the back-end to a larger server or multiple instances)if user demand grows. Furthermore,updates to the model (say a newer version of the EfficientNet or an entirely new model) can be deployed to the server without requiring any change on the user's side,since the front-end always fetches results from the server's current model.This makes SonaDx a flexible platform for continuous improvement.

In summary,the technical stack of SonaDx reflects a balance between innovation and practicality.The use of standard web technologies (React,Flask) aligns with what a skilled undergraduate team can implement, while the integration of the optimized AI model

demonstrates how cutting edge deep learning can be delivered through a well designed software system.The result is a robust application where each layer front end, back end, and cloud deployment works in concert to provide a seamless diagnostic tool to the end user.

6.5 Use Case Diagram Description

The use case diagram for SonaDx illustrates how users interact with the system and what functionalities the system provides in response.In this diagram,we identify two main actors and their interactions:

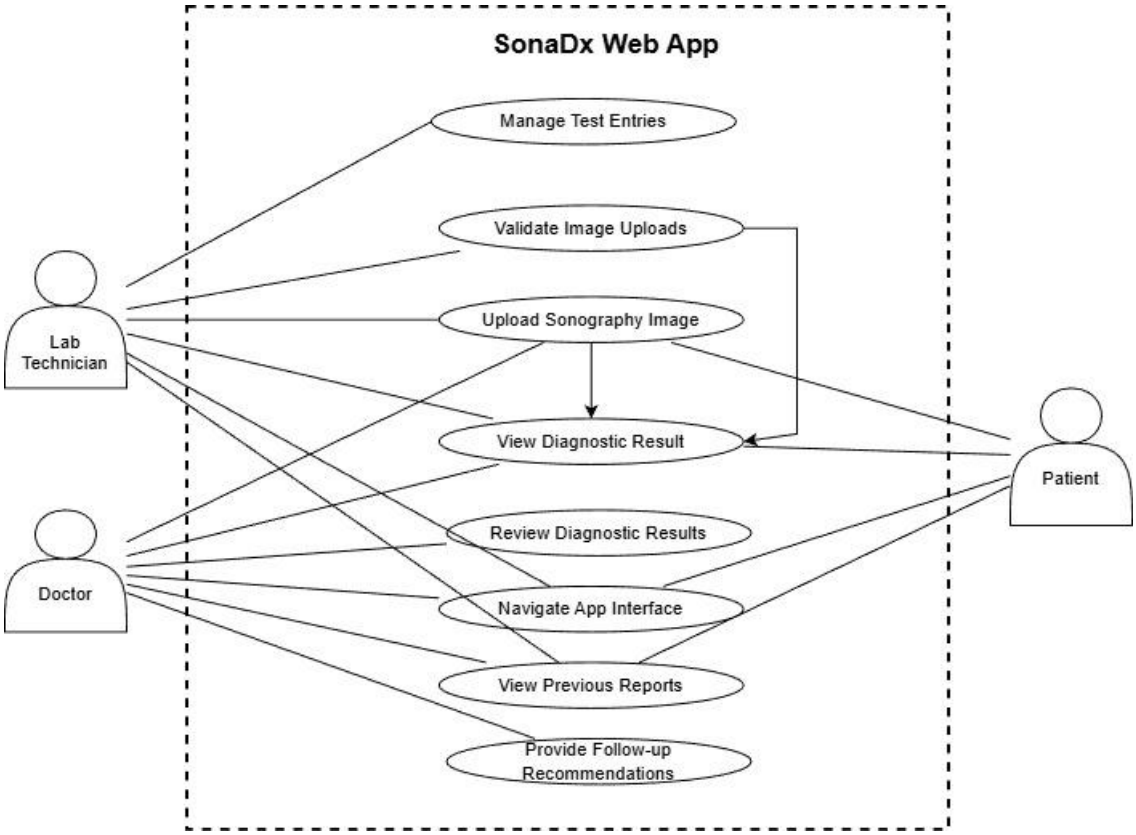


Figure 12 : Use case Diagram

- **User (Actor):**The primary actor in the use case diagram is the User,who in the context of SonaDx could be a healthcare professional (such as a radiologist, sonographer, general physician)or a trained technician,or even a patient in a telemedicine scenario.Essentially, anyone who has an ultrasound image and seeks a diagnostic assessment can be considered a User.This actor is external to the system but initiates all the use cases.
- **SonaDx System:**Represented as a system boundary in the use case diagram, the SonaDx application itself is the second “actor”(though it’s an automated system rather than a person).It encompasses the software components the web interface and the backend AI service that will carry out the use cases triggered by the User.

Within the diagram, several use cases (functionalities)are shown as oval shapes, and the User is connected to each of these use cases to indicate they can initiate them.The key use cases include:

- **Upload Ultrasound Image:**This is the fundamental use case where the User provides an input image to the system.The User triggers this by selecting a file and confirming the upload.In the diagram,the User actor has an association with this "Upload Image" use case,meaning the user has the ability to perform this action.
- **Request Analysis:**Once an image is uploaded, the User effectively requests the system to analyze it (sometimes this is implicit right after upload).This use case represents the interaction where the User asks the system for a diagnosis on the provided image.In some diagrams, this might be combined with the upload step, but it can be considered a distinct use case the user's goal is to get an analysis result. The SonaDx system will handle this request internally by running the AI model (this internal processing can be represented as an included or automated sub-process,but from the user's perspective, it's part of the "Analyze Image" use case).
- **View Diagnostic Result:** After processing, the system delivers a result back to the User.The User then engages in the "View Result" use case,where they see the output on the screen.The diagram would show the User connected to this use case, indicating the user's ability to obtain and view the diagnostic outcomes (classification of the image,and possibly the heatmap visualization).
- **Navigate App/Other Interactions:** A more general use case can be "Navigate Interface" or "Access Information,"encompassing things like the user viewing the home page, reading instructions, or navigating to an About section.While not as critical as the core diagnostic flow, it's still a possible interaction: the User can use the navigation menu to move between pages (Home, Upload, Results).This could be represented in the diagram if needed for completeness, showing that the user can, for instance, "View App Info."

The relationships between these use cases can also be described.For example, "Request Analysis"might include the sub-process of System Processes Image, which is an internal function of the SonaDx system not directly triggered by user action but automatically happening as part of the analysis. Similarly,"View Diagnostic Result"extends from the successful completion of the analysis.In the use case diagram, such internal details are usually omitted or shown as notes, to keep the diagram user-centric.The main focus is that the User uploads an image and receives a result.

The roles are clearly delineated:the User is the actor initiating actions, and the System is the one performing the analysis and providing feedback.There are no other human actors (such as an administrator) in the basic SonaDx use case scenario,since the application is designed to be fully automated and user-driven.All the heavy work (image interpretation) is handled by the SonaDx system behind the scenes once the user starts the process.

In summary, the use case diagram for SonaDx would show a simple and direct interaction flow: Upload Image → Analyze Image → View Result, with the User at the left initiating these and the SonaDx system on the right providing the functionality.This textual description

of the use case diagram underlines how the app is used and what roles are involved, giving a clear picture of the system's functional requirements from a user's perspective.

6.6 App Workflow Diagram Description

The app's workflow can be described as a sequence of steps that a user and the system go through from start to finish.This workflow is often depicted in a flowchart or activity diagram,but here we will outline it in a step-by-step narrative form from the user's perspective:

- **Launching the Application:**The user opens a web browser on their device and navigates to the SonaDx web application(for instance, by entering the provided URL). The SonaDx home page loads, presenting the user with the title and a brief introduction or welcome message. On this home screen, the user may see an option to start analysis or an "Upload Image"button prominently displayed, as well as basic navigation links(to other sections like About or Help if available).At this point, the system is idle, waiting for user input.

- **Navigating to the Upload Interface:**The user proceeds to the analysis function. Depending on the app design, this could mean clicking on a "Start Diagnosis" or "Upload Image" call-to-action on the home page.The interface then shows the upload form (if not already present on the home page).This page or section typically contains a file selection button (often labeled something like "Choose Ultrasound Image") and perhaps instructions or tips (e.g, "Please upload a clear ultrasound image of the breast in PNG or JPG format.").The user is now ready to provide the input data to the system.

- **Selecting and Uploading an Image:** The user clicks on the file selection button, which opens a file chooser dialog (or prompts them to use their device's camera, if the app supports taking a photo of an ultrasound printout a possible use case for mobile users).The user selects the ultrasound image file from their device. The filename may then appear on the interface, confirming the selection. Next, the user submits the image for analysis this could be an automatic upload as soon as the file is chosen, or the user might need to click an "Analyze"or "Upload" button to confirm. At this moment,the application starts uploading the image data to the server. The user's action triggers an API call from the front-end to the back-end (Flask API),sending the image file for processing. The interface likely provides feedback that the image is being uploaded and analyzed, for example by displaying a loading spinner or a message like "Analyzing image, please wait…".

- **Backend Processing and Analysis (System Action):**Once the image reaches the server, the SonaDx back-end carries out the analysis (this step is mostly invisible to the user aside from the loading indicator). The server loads the image, pre processes it, and runs it through the EfficientNetB3 TFLite model.The model generates a prediction of the class (Normal, Benign, or Malignant).Simultaneously, the system might also generate the Grad CAM heatmap based on the model's result to highlight

important features.The back-end then prepares a response containing the outcome.If any error occurs (e.g., the image is not readable or the server is down), the back-end would send an error message instead. In the normal flow, this computation only takes a brief moment.

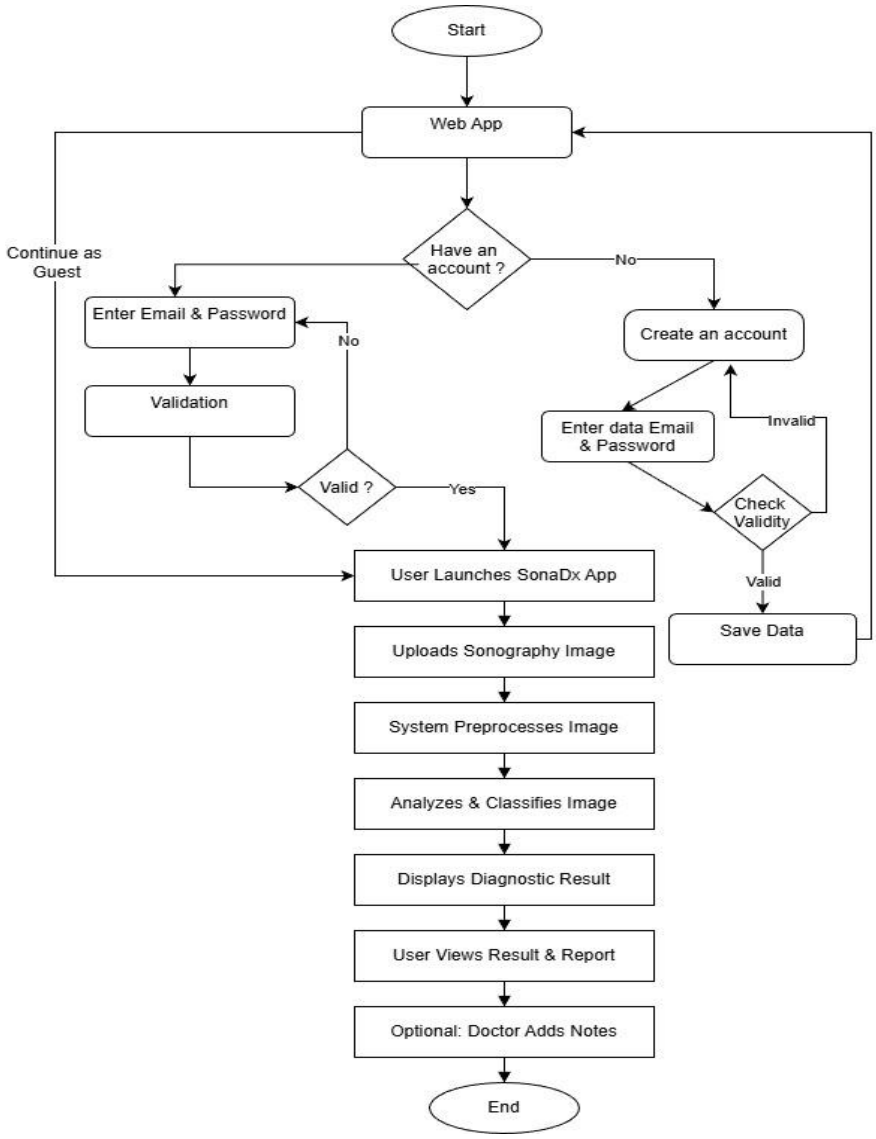


Figure 13: App Workflow Diagram.

- **Receiving and Displaying Results:** The front end continuously waits for the response from the back-end.As soon as the analysis is done, the result is sent back to the front-end in a structured format (for example, a JSON response that includes the predicted class label and possibly a probability/confidence score, along with the heatmap image encoded in a suitable format).The SonaDx front-end receives this data and then automatically transitions the user interface to the Results view. In this results page/section, the user sees the outcome of the analysis.For instance, the screen might

show a big, highlighted label like “Malignant Detected” or “Benign,” etc., possibly accompanied by a short explanatory text (“The uploaded ultrasound shows characteristics of a malignant tumor.”).Right beside this, the app displays the uploaded ultrasound image with a colored overlay (the Grad-CAM heatmap)indicating where the model focused its attention.The result page may also show additional info such as the confidence level or a legend explaining the heatmap colors (e.g., red areas indicate higher importance).If the result is Normal, the display might reassure the user that no signs of a tumor were detected in the image.

- **User Interprets the Results:** The user now examines the output.If the user is a clinician, they might compare the AI’s highlighted regions with their own mental notes of the image.If the user is not a specialist, they may rely on the color-coded result and any provided guidance (for example, the app might advise something like “Note: This is not a definitive diagnosis. Please consult a doctor for confirmation.” if it’s designed to include such caution).The user can use the results to inform their next steps for example, scheduling further tests for a malignant suggestion, or feeling relief and continuing routine screenings if the result was normal (always with the understanding that clinical correlation is needed).The interpretability provided by the heatmap makes this step more engaging and informative for the user.

- **Post-Analysis Options:** After viewing the results, the user typically has a few options in the workflow.They might choose to save or download the results (if the app provides a feature to download a report or screenshot).They might also want to analyze another image for instance, a clinician could have multiple images from the same patient or different patients. The interface therefore provides a button like “Analyze Another Image” or “Start Over.” By clicking that,the user can return to the upload step and repeat the process with a new image. Alternatively, the user could navigate to other parts of the app (like an About page to learn more about SonaDx, or perhaps a Feedback form if present).If the user is finished, they simply close the application or browser tab.

- **Closing the Application:** This final step is the end of the session.Since SonaDx is a web app, there’s no formal “logout” needed (as there are no user accounts in the current version).The user can just exit.All transient data (the uploaded image, the result) is not stored on the server, so once the session is over, no lingering patient data remains in the system.If the user later returns to the app, the process begins anew.

This workflow ensures that from start to finish, using SonaDx is straightforward and mirrors the typical clinical process:provide test input (imaging) → get analysis → consider next steps.The app’s design abstracts away the complexity, so the user experiences a smooth journey through these steps with clear guidance at each juncture.The App Workflow Diagram that accompanies this description would depict these steps in order, showing the user’s actions and the system’s responses in a flowchart style,thereby validating that the application logic covers all necessary stages from image input to results output.

6.7 App Interface Demonstration

The user interface (UI) of SonaDx was carefully crafted to be intuitive and informative.In this section, we describe what a user sees and how they interact with the app's UI components on each page or screen.The goal of the interface design is to make the diagnostic process as simple as possible, even for users who may not be very tech savvy.

Home Page: When users first access the SonaDx application, they are greeted by a clean home page.This page likely displays the application title "SonaDx Early Breast Cancer Detection from Sonography Images" prominently, along with a brief description or tagline such as "Upload a breast ultrasound image to get an instant diagnostic assessment." The styling is professional yet approachable for instance, a calming color scheme (perhaps blues or greens) that is easy on the eyes.There may be a representative icon or logo for SonaDx (for example, a stylized image of a breast or an ultrasound icon) to give a visual identity.The home page provides basic navigation; at the top, a header bar could have the SonaDx logo on the left and simple menu options on the right (for example,Home, Analyze or Upload, and About).If an "Upload" button is present on the home page itself, it is highlighted to invite the user to start the process immediately. Overall, the first impression is that of a modern web app with a straightforward purpose, with no clutter or unnecessary content.

Upload/Analysis Page: The core interaction happens on the upload page. Here, the layout is focused on a central panel or box where the user can select an ultrasound image.The design follows standard conventions: a large "Upload Image" button or a drag-and drop target area is provided. The instructions might say,"Click or drag an ultrasound image here to upload for analysis."There might also be an illustrative icon (like a cloud upload symbol) to reinforce what needs to be done. For users on mobile devices, tapping the upload area might offer them the choice to take a photo using their camera or select an existing image from their gallery the interface is designed to handle both scenarios gracefully.Once an image is selected, a small preview thumbnail could be shown, confirming the correct file is chosen. The user then clicks an "Analyze" or "Submit" button to initiate the process if it's not set to auto-run.At this stage, the interface gives feedback that the image is being uploaded and processed. For example, a loading spinner or progress bar appears. The app might display a message such as "Analyzing please wait a moment."During this short wait, the user is not left guessing – the UI indicates that work is in progress.The simplicity of this page ensures that the primary action (uploading an image) is obvious and there are no distracting elements.If there are any constraints (like image file size or format requirements),those are mentioned in a friendly note below the upload button.

Results Page: After analysis,the UI automatically transitions to the results page. This page is arguably the most important from a user's perspective, as it conveys the diagnostic findings. The design presents the information in a clear, prioritized manner. At the top, a headline might read "Analysis Result" or"Diagnosis". Just below, the outcome is shown – for instance, a large text could say "Malignant Detected" or "Benign – Likely Fibroadenoma" or"No Tumor Detected (Normal)" depending on the model's prediction. This text is often styled in a distinctive way: possibly color-coded (red text for malignant, orange for benign, green for normal) and maybe accompanied by an icon (a checkmark or warning symbol) to reinforce

the result category. Directly beneath the textual result, the app displays the ultrasound image with the Grad-CAM heatmap overlay.The image is usually shown in a container that allows the user to see the details; if the user clicks or taps the image, perhaps it can enlarge or open in a modal for closer inspection (this is an optional enhancement that may or may not be present).A legend or caption is provided next to the image explaining that the colored regions indicate areas that influenced the AI's decision. For example, a small note might say "Red areas on the image highlight where the system detected the most significant features."The combination of text and visual makes it easy for the user to interpret the findings they can correlate the highlighted regions with anatomical structures in the ultrasound.

On the results page,there may also be additional information or guidance. Since this is a medical application, a disclaimer or note is often included, such as: "This result is an aid and not a definitive medical diagnosis.Please consult a healthcare professional for confirmation." This reminds users of the app's supportive role. If the result is malignant, the app might advise something like "Consider follow up with a specialist as soon as possible."If benign or normal, it may still suggest routine monitoring.These pieces of text are written in a reassuring and clear tone to help users understand the implications without causing undue alarm or false security.

The results page also features navigation elements for what to do next.Prominently, there is a button like "Analyze Another Image" which brings the user back to the upload interface to perform another analysis.This button might appear as a bright, inviting element near the bottom of the results section. Additionally, a "Back to Home" link or button could be present if the user wants to return to the introduction page. The navigation bar at the top remains available,so the user can always click "About" to read more about the project or any other section provided.

About/Info Page: If the application includes an "About"page or similar, clicking that link would take the user to a section describing the SonaDx project, the team, or how the AI works (in non-technical terms).The UI here would be text centric, possibly listing the motivation behind SonaDx, the accuracy achieved (for example, mentioning the 94.5% test accuracy of the model), and acknowledgments to the supporting organizations or datasets (like BUSI etc., if the developers chose to mention them).This page would be written in a user-friendly way, avoiding heavy jargon,since its purpose is to inform interested users (like doctors or stakeholders) about the app.

Responsive and Adaptive UI: Throughout all these pages, the UI is designed to be responsive. In demonstration, if we shrink the browser window or access it on a mobile phone, the layout adapts.On a desktop, we might see a multi-column layout (for instance, result text on the left and image on the right on the results page).On a mobile screen, these could stack vertically: first the result text, then the image below it, both centered and sized appropriately. The navigation menu might collapse into a mobile "hamburger" menu icon on a small screen to save space. Buttons become larger or more touch friendly on mobile. The demonstration of the interface on different devices shows consistent functionality and ease of use users do not have to pinch zoom or struggle with tiny text on mobile, because the design inherently accommodates different screen sizes.

Visual Design Considerations: In terms of visual aesthetics, the app demonstration would show a consistent color theme and typography.SonaDx likely uses a sans-serif font for a clean, modern look, with larger font sizes for headings and legible sizes for body text. Icons (if used) are simple and universally understood. Contrast is sufficient to read text easily (important for accessibility). Additionally, the design avoids overwhelming the user with too much information at once each page focuses on one primary action or piece of information.This follows good UI/UX principles, especially crucial in a healthcare app where users might be in stressful situations; they need clarity and straightforward interactions.

In conclusion,the app's interface is demonstrated to be user centric: everything from the layout, wording, to the colors is chosen to make the experience pleasant and clear.The interface essentially walks the user through the process step by step, and even someone with basic computing skills could use SonaDx to get a diagnostic reading from an ultrasound image with minimal assistance.The successful implementation of this UI is evidenced by how smoothly one can navigate from the home page to getting a result, reflecting the project's emphasis on usability.

6.8 Practical Benefits

The SonaDx application offers numerous practical benefits in a healthcare and diagnostic context, particularly for communities that have traditionally been underserved by specialist medical services.By transforming a complex AI model into a handy tool,SonaDx demonstrates tangible advantages that go beyond the laboratory and into the real world of patient care:

Expanded Access to Early Detection: One of the foremost benefits of SonaDx is the expansion of access to early breast cancer detection.In many rural or low resource areas, access to expert radiologists or oncologists is limited.Patients often have to travel long distances to urban centers for specialist consultations, which can lead to delays in diagnosis.With SonaDx, any clinic equipped with an ultrasound machine and basic internet can perform an on the spot analysis of ultrasound images.This means that a woman in a village can get a preliminary screening result immediately after her ultrasound scan, rather than waiting weeks for an expert to review the image. Early detection is critical in breast cancer outcomes;catching a malignant tumor earlier can significantly increase survival rates. SonaDx thus serves as a frontline screening assistant,helping flag suspicious cases that need urgent follow up.

Supporting Healthcare Workers in Remote Areas: In remote clinics or community health centers, it's common that general physicians or technicians with limited specialization handle a wide range of tasks.SonaDx empowers these healthcare workers by giving them a second pair of eyes, so to speak. For instance, a general doctor who is not highly experienced in interpreting breast ultrasounds can use SonaDx to double check their own assessments.The AI might catch subtle patterns that the doctor could miss, or conversely, confirm the doctor's suspicion, thereby boosting confidence.In places where the radiology workforce is small and declining[54], such a tool can be invaluable.It effectively extends the reach of radiology expertise to places where no radiologist is available on site.This not only improves diagnostic

accuracy in those settings but also helps reduce the workload and burnout of the few specialists by filtering cases (only the complex or high risk cases might then be referred onward).

Cost and Time Savings for Patients: By providing immediate results, SonaDx can save patients both money and time. Take the example from Bangladesh:government hospitals charge a minimal fee for ultrasound, but due to lack of radiologists,patients often had to go to private centers and pay much more for results[55].With a tool like SonaDx in public clinics, patients can receive a diagnostic assessment during their visit without additional costs. Even though SonaDx is not a complete replacement for a pathology-confirmed diagnosis, it can guide decisions such as whether a patient should be referred for a biopsy or a mammogram.This triaging can prevent unnecessary expensive tests for those likely normal, and conversely, expedite referrals for those who truly need them.The overall healthcare system benefits when resources are focused correctly, and patients benefit by avoiding both the anxiety of long waits and the expense of seeking multiple opinions.

Consistency and Reliability: Human interpretation of ultrasound images can be subjective and prone to variability.Fatigue, experience level, and cognitive bias can affect readings. SonaDx provides a consistent level of analysis every time.While not infallible, the AI model was trained on a large dataset and provides standardized outputs based on learned patterns. In practice, this means it could reduce the incidence of oversight.For example, an overworked technician might overlook a faint shadow on an image, but the AI could still pick it up as an abnormal pattern.Using SonaDx in parallel with human judgement thus adds a safety net a practical benefit in busy clinics with high patient loads.

Reaching Underserved Populations: Because SonaDx is a web-based tool, it can be accessed anywhere.This includes not just rural clinics but also mobile health units, refugee camps, or temporary screening camps in remote areas. All that is needed is a portable ultrasound device (which are increasingly available and affordable) and a smartphone or laptop. The combination of portable ultrasound + SonaDx effectively creates a mobile diagnostic clinic. For populations that cannot easily reach urban hospitals due to distance, cost, or political instability this can bring life-saving diagnostics to their doorstep. It aligns with global health initiatives of using digital health to overcome geographic and socioeconomic barriers.

Integration into Healthcare Systems: From a practical point of view, SonaDx can also integrate with telemedicine workflows. A patient in a rural clinic could have their ultrasound done locally, the images analyzed by SonaDx on site, and then the results could be shared with a specialist in the city through telemedicine platforms for further advice.This synergy means better collaboration between primary care and specialized care. The presence of an initial AI result can make teleconsultations more focused and efficient (the specialist can agree or disagree with the AI and quickly advise the next step).

In summary,the practical benefits of SonaDx are seen on multiple levels: for patients, it offers quicker,cheaper preliminary diagnostics; for healthcare providers,it offers decision support and extends expertise;for health systems, it optimizes resource use and extends services to hard to reach areas.Ultimately,by improving early detection rates, SonaDx contributes to

better clinical outcomes catching breast cancer cases earlier when treatment is more likely to succeed, and thereby potentially improving survival and quality of life for patients who would otherwise be diagnosed late.The app is a clear example of digital technology contributing to more accessible and equitable healthcare, in line with global health objectives.

6.9 Future Plans

While the current version of SonaDx is fully functional and achieves its primary goals,the development team has identified several areas for future enhancement.These planned features and improvements aim to broaden the app's scope,increase its usability, and ensure it remains at the cutting edge of healthcare technology:

- **Multi-Disease and Multi-Modal Detection :** One future direction is to extend SonaDx beyond early breast cancer detection.The underlying platform could be adapted to detect multiple diseases or conditions from medical images. For example, the team envisions training additional models for other types of ultrasound diagnostics such as thyroid nodules, liver lesions, or obstetric ultrasounds and integrating them into the app.SonaDx could evolve into a suite of diagnostic tools, where a user selects the type of screening (breast,thyroid, etc.) and uploads the relevant image. Moreover, integrating other imaging modalities is also a goal.In the context of breast cancer, a future version might accept mammography images or even MRI scans, combining their analysis with ultrasound results (a concept known as multi-modal diagnosis). This could improve accuracy and allow the app to serve as a one-stop solution for breast imaging analysis.

- **Local Language Support:** To better serve users in Bangladesh and other non English speaking regions, future iterations of SonaDx will include multi language support, especially Bengali (Bangla).This means the user interface text, instructions, and result outputs would be available in the local language, making it easier for local healthcare workers and patients to use.Implementing language toggle options or automatically detecting the user's locale can greatly increase the app's adoption. Along with language, cultural localization will make SonaDx feel more tailored to its target communities.

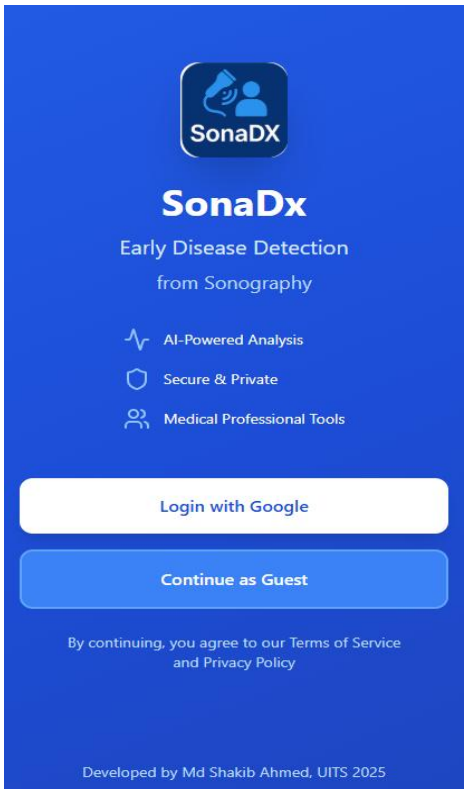
- **Offline and Mobile App Version:** Recognizing that internet connectivity can be unreliable or absent in some rural areas,a major future plan is to develop an offline-capable mobile application for SonaDx.This would likely involve creating an Android app (and possibly iOS in the future) that embeds the TensorFlow Lite model on the device. By doing so,the analysis could be done entirely on a smartphone or a tablet without needing to send data to a server. An offline mode would be transformative for health camps in remote villages or during disaster relief scenarios where connectivity is limited.The team has suggested an Android app using the same TFLite model as a next step,which could even leverage device hardware like Neural Processing Units (NPUs) for faster inference on phones. This mobile app could cache results and then sync with a central database when back online, combining the best of offline functionality and online record keeping.

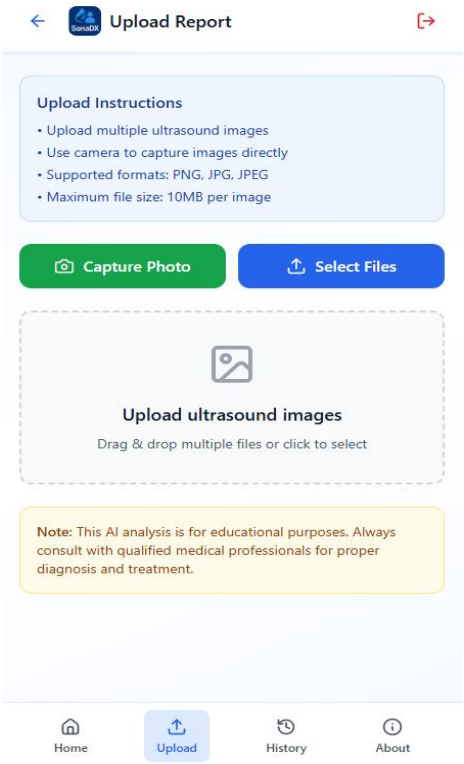
- **Enhanced Security & Privacy:** As usage scales up, future work will ensure the app complies with healthcare data protection standards (like HIPAA or GDPR where applicable). This may involve adding features like end-to-end encryption, patient consent modules, or even exploring federated learning so that the model can improve using data from different hospitals without those images ever leaving the hospital's control. Though behind-the-scenes from a user perspective, these improvements will increase trust among institutional users.

- **Continuous Model Upgrades:** As deep learning technology advances, SonaDx will keep pace by evaluating newer architectures or techniques. Future plans include testing models like EfficientNetV2 or vision transformers (ViT) for potentially improved accuracy. Any superior model can be integrated into the app's backend as long as it meets the speed and size requirements for deployment. The modular design of the system allows such swaps without changing the user experience.

All these future plans center on a common theme: making SonaDx more powerful, user-friendly, and widely applicable, while maintaining reliability. The aspiration is that in a few years, SonaDx could evolve from a focused breast cancer screening tool to a comprehensive diagnostic aid platform used in many clinics.Each planned feature will undergo the same rigorous approach of development and testing as the current version,to ensure that added complexity truly translates into added value for users.By outlining these plans,the team acknowledges that SonaDx is not an end but rather a platform that can continuously grow,keeping the project relevant and useful in the fast-changing landscape of digital health.

6.10 Picture of the UI





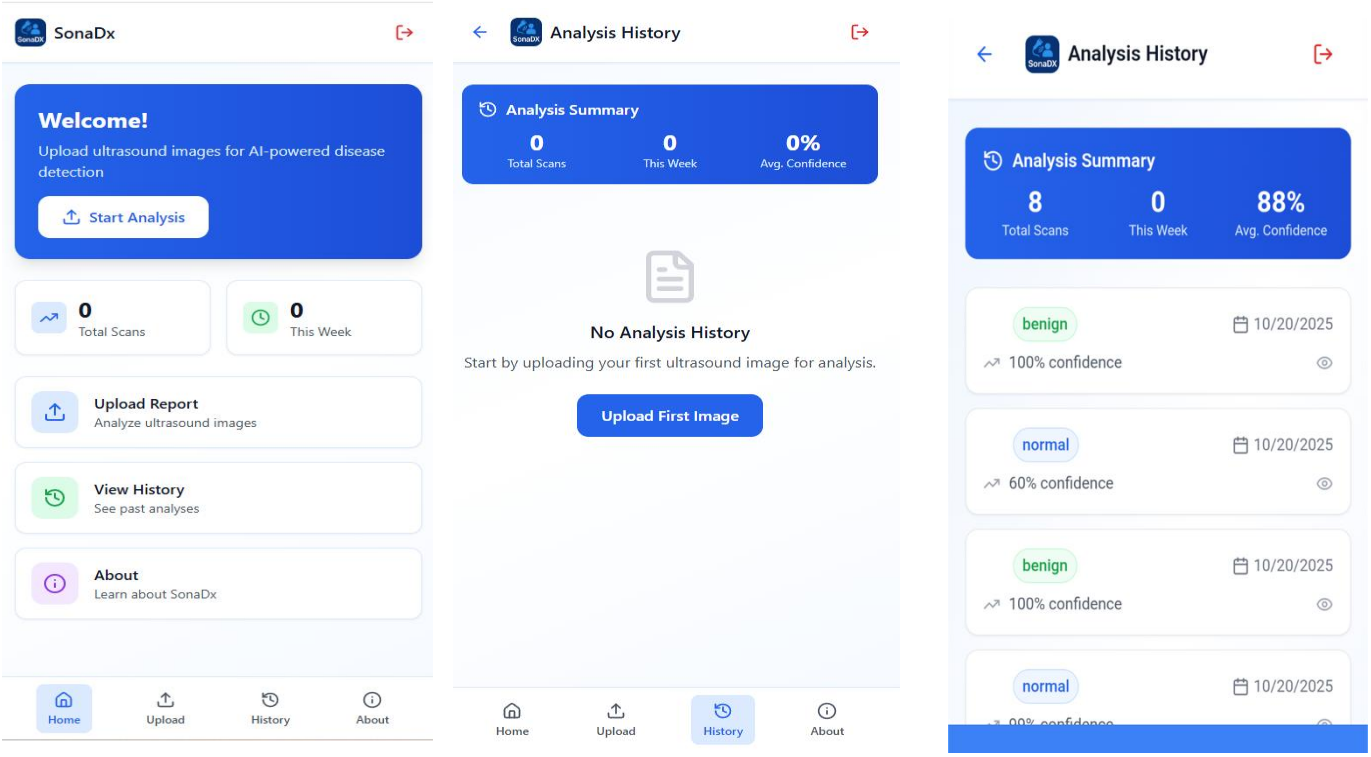


Figure 14: UI design of the webapp

6.11 Conclusion

In conclusion,the development of the SonaDx mobile/web application stands as a significant achievement in this capstone project, embodying the translation of advanced technology into an accessible healthcare tool.Chapter 6 has detailed how the app was conceived and designed to meet real-world needs from its straightforward user interface to the robust technical infrastructure under the hood. By successfully implementing image upload, AI analysis, and result visualization in a cohesive platform, the team has demonstrated how a research prototype (the breast cancer detection model)can be transformed into a practical solution that people can use in everyday clinical scenarios.

The SonaDx app contributes to digital healthcare accessibility by breaking down several barriers. Geographical barriers are reduced because any clinic with basic internet can use the app to get diagnostic insights, as opposed to relying solely on distant experts. Knowledge barriers are addressed by the app's user friendly design and explainable outputs, which allow even non-specialists to benefit from and trust the AI's assistance. Economic barriers are also considered the app is deployed on readily available infrastructure and free for end-users, aligning with the goal of low cost interventions in public health.

Moreover, SonaDx exemplifies the power of interdisciplinary integration:it combines computer science (web/mobile development, cloud computing) with data science (machine learning models) and domain-specific knowledge (medical imaging) to create a tool greater than the sum of its parts.For an undergraduate project, this is a notable demonstration of leveraging classroom learning to solve a pressing societal problem.The app not only

showcases the team's engineering skills but also their understanding of user needs and healthcare context, which is crucial for any technology aiming to make a real impact.

As it stands,SonaDx is a pioneering step towards smarter healthcare in Bangladesh and similar contexts.It shows how early detection of diseases like breast cancer can be aided by technology, potentially leading to earlier treatments and improved patient outcomes. The project, however, doesn't end here as outlined, there are rich possibilities for future enhancements, from expanding the diagnostic scope to deploying on mobile devices and integrating local languages. Each of these steps will further cement SonaDx's role in making healthcare more inclusive and proactive.

In summary, the successful design and implementation of the SonaDx application reinforces the thesis's central message: that modern computing solutions can bridge critical gaps in healthcare delivery. By bringing an AI-driven diagnostic tool to the fingertips of those who need it most, SonaDx contributes to a future of intelligent healthcare where technology and human expertise collaborate seamlessly for faster, more accurate, and more accessible medical diagnostics. The capstone journey concludes with a working application that not only underscores the technical accomplishments of the team but also holds promise for meaningful real world use a fitting culmination of an academic endeavor aimed at creating positive societal impact.

Chapter 7

Conclusion & Future work

7.1 Conclusion

SonaDx, a deep learning-based detection system for breast cancer using ultrasound images, was developed and implemented successfully in this study. The power of the EfficientNetB2 Convolutional Neural Network (CNN) was used in this system which directly leverages fine-tuning from a hybrid dataset made up of 2,644 breast ultrasound photographs from BUSI, MT_Small as well as Double-Test-Set examples for Kaggle and local Bangladeshi samples.

This research has shown that deep learning, once well trained and refined, can achieve high performance in breast tumor diagnosis and classification. The final model achieves a testing accuracy of 94.5%, which shows excellent generalization power under various imaging conditions.

Through systematic preprocessing, augmentation and fine-tuning, the model effectively classified Normal, Benign, and Malignant lesions. By means of TensorFlow Lite conversion it could be made lightweight for web and mobile platforms; and with Grad-CAM visualization improved interpretability was provided [56] . A medical doctor can use this to monitor what areas influenced the AI's choice from visual information directly.

Also, SonaDx was deployed through Flask API hosted on Hugging Face Spaces and integrated into a modern React-based web application, achieving real-time AI inference in clinical or educational setting.

In effect, SonaDx provides a bridge between academic AI research and real-world healthcare by delivering an accessible, understandable, and efficient capitalist diagnostic tool tailored for Bangladesh and its low resource counterparts .

7.2 Future Works

Although the proposed SonaDx system demonstrates promising performance, several key directions remain for future enhancement:

1. **Dataset Expansion and Clinical Validation**
   Future work should focus on expanding the dataset by incorporating ultrasound images from multiple Bangladeshi and international hospitals. In parallel, real world clinical validation under actual hospital workflows is necessary to assess generalizability, robustness, and clinician acceptance [57].
2. **Multi-Modal Diagnostic Framework**
   Integrating breast ultrasound with complementary imaging modalities such as mammography or MRI can improve diagnostic accuracy, particularly for complex or ambiguous cases, leading to a more comprehensive decision-support system [58].
3. **Model Optimization and Advanced Architectures**
   Exploring newer architectures such as EfficientNetV2 and Vision Transformers (ViT) can further optimize the trade off between accuracy, computational efficiency, and

inference speed, making the system more suitable for deployment on resource-constrained devices.

4. **Explainability and Trust Enhancement**
   Incorporating advanced interpretability techniques, including attention-based visualization or Layer wise Relevance Propagation (LRP),can enhance transparency and clinician trust by providing clearer insights into model decision-making [59].
5. **Deployment and Accessibility**
   Developing an offline-capable mobile application using TensorFlow Lite would enable real-time diagnosis in rural or low-connectivity environments, significantly increasing the system's practical impact [60].

These future improvements can transform SonaDx from a research prototype into a clinically deployable, scalable, and trustworthy AI diagnostic assistant.

7.3 Final Thought

The development of SonaDx represents the potential for artificial intelligence to change the scene in medical diagnostics. That is especially true in countries like Bangladesh where resources are scarce. Using deep learning, current web technologies, and a well-chosen dataset geared specifically to the region, this project teaches us about the role AI could have in supporting medical professionals in providing diagnostic services for under-resourced areas. Problems like the diversity of data, contradictory opinions from clinical outcomes, privacy as well as other issues continue to be sources of headache for this work. Nonetheless, it provides a sound basis which future generations of researchers can build on. The findings of the study showed that the use of AI for breast cancer diagnosis is not only possible technically but also has direct practical value for early diagnosis and saving people's lives. Still, the development of SonaDx helps point the way forward for intelligent Healthcare. Bio-technology and human acumen working together mean faster right diagnosis.

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